Bauhaus-Universität Weimar

Physics-Informed Neural Networks for Multi-Axis Structural System Identification

A Direction-Agnostic Framework for Stiffness and Damping Recovery from Single-Node Displacement Records

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Abstract

The inverse identification of structural parameters from measured dynamic response is a foundational problem in structural health monitoring, model updating, and digital-twin calibration. Classical inverse solvers — least-squares regression on numerically differentiated response, subspace identification, and frequency-domain modal fitting — are effective but inherit a structural weakness: they require explicit estimates of velocity and acceleration, which must be obtained by differentiating a sampled displacement record. Differentiation amplifies high-frequency content, and the resulting estimator is sensitive to sampling rate, endpoint treatment, and the spectral content of the excitation.

This chapter develops and evaluates a physics-informed neural network (PINN) formulation that removes this dependency. A fully connected network with Fourier feature encoding represents the displacement field $u(t)$ as a continuous, infinitely differentiable surrogate. Velocity and acceleration are obtained by exact automatic differentiation of that surrogate rather than by finite differencing of the data, and the governing equation of motion is imposed as a residual penalty at collocation points distributed throughout the observation window. The unknown stiffness K and damping C enter as trainable scalar parameters, optimised jointly with the network weights against a composite loss. Only displacement is supplied during training; neither velocity, acceleration, nor any prior estimate of K or C enters the data term.

The framework is implemented as a direction-agnostic pipeline over a finite element model of a steel support table (total structural mass 78,500 kg), driven in a transient analysis at 15 Hz and observed at a single node (Node 3575). A single YAML-configured code path, with no modification to the network, the loss, or the optimiser, was executed independently in the X, Y, and Z directions. The same architecture converged in all three directions.

Stiffness was recovered accurately in every direction. Against an independent least-squares estimator evaluated under a consistent force convention, the identified stiffness agreed to within 0.28 % in X, 0.38 % in Y, and 0.015 % in Z. The displacement surrogate generalised to held-out samples never presented to the network, achieving coefficients of determination of 0.99992, 0.99989, and 0.99999 respectively. Damping was recovered in the X and Y directions, where the identified coefficients correspond to modal damping ratios of 0.250 % and 0.242 %. In the Z direction, stiffness converged to a final-window stability of 0.012 % while the damping coefficient retained a 24.4 % range over the final 5,000 iterations; the Z damping coefficient is therefore reported as constrained but not converged over the available 10-second record, for reasons developed from the decay-envelope analysis in Chapter 5.

A supporting window-length study establishes the governing physical principle behind this behaviour: stiffness is identifiable from a small number of oscillation cycles, because it is encoded in the instantaneous curvature-to-displacement ratio, whereas damping is identifiable only from the decay of the free-response envelope and therefore requires an observation window commensurate with the decay time constant $\tau = 2M/C$. The framework's capacity to distinguish these two regimes — returning a tightly converged stiffness alongside a damping

coefficient whose convergence diagnostics reflect the information actually present in the record — is presented as a substantive methodological result rather than a limitation.

Keywords: physics-informed neural networks, structural system identification, inverse problems, Fourier features, spectral bias, DeepXDE, modal reduction, automatic differentiation

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Chapter 1

Introduction

1.1 Motivation: The Inverse Problem in Structural Health Monitoring

The forward problem of structural dynamics is, in a formal sense, settled. Given a spatial discretisation of a structure, a constitutive law, a set of boundary conditions, and a prescribed excitation, the response follows from the numerical integration of a system of ordinary differential equations. Commercial finite element codes execute this calculation routinely and to high accuracy. The difficulty that occupies structural engineering practice is the converse: given a measured response, what are the physical properties of the structure that produced it?

This inverse question is the operational core of structural health monitoring. A structure in service departs from its design model. Connections loosen, supports settle, materials degrade, added mass accumulates, and the boundary conditions assumed at the design stage are rarely those realised in the field. The stiffness matrix of the as-built structure is therefore not the stiffness matrix of the design model, and the difference between them is precisely the quantity of engineering interest. A loss of stiffness localised to a region is the signature of damage; a global shift is the signature of a changed support condition. Recovering these quantities from response measurements is what permits a finite element model to be updated, a digital twin to be calibrated, and a condition assessment to be made on evidence rather than assumption.

The measurement that is most readily and reliably available in practice is displacement, or a quantity from which displacement is directly obtained. Accelerometers are inexpensive and widely deployed; displacement transducers, laser vibrometers, and increasingly vision-based tracking systems provide direct kinematic records. What is almost never available is a direct measurement of the internal force distribution, and it is the internal force that carries the stiffness and damping information. The inverse problem is therefore one of inferring a force-related quantity from a kinematic one, mediated by the equation of motion.

1.2 Limitations of Classical Inverse Solvers

The classical approach to this inference is direct. For a single-degree-of-freedom idealisation governed by

$$M\ddot{u}(t) + C\dot{u}(t) + Ku(t) = F(t), \tag{1.1}$$

the equation is linear in the unknown parameters K and C. Rearranging,

$$C\dot{u}(t) + Ku(t) = F(t) - M\ddot{u}(t), \tag{1.2}$$

and evaluating at N sample times $t_1, \ldots, t_N$ yields an overdetermined linear system

$$\mathbf{A}\boldsymbol{\beta} = \mathbf{b}, \qquad \mathbf{A} = \begin{bmatrix} \dot{u}(t_1) & u(t_1) \\ \vdots & \vdots \\ \dot{u}(t_N) & u(t_N) \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} C \\ K \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} F(t_1) - M\ddot{u}(t_1) \\ \vdots \\ F(t_N) - M\ddot{u}(t_N) \end{bmatrix}, \tag{1.3}$$

whose least-squares solution is $\hat{\boldsymbol{\beta}} = (\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}\mathbf{A}^\mathsf{T}\mathbf{b}$. This estimator is unbiased under the assumption of independent, homoscedastic residuals and an adequate model, and it is computationally trivial. It is the natural baseline against which any more elaborate method must justify itself, and it is used in that role throughout this work.

Its weakness is structural rather than incidental. The design matrix $\mathbf{A}$ requires $\dot{u}$, and the right-hand side $\mathbf{b}$ requires $\ddot{u}$. Neither is measured. Both must be constructed from the sampled displacement record by numerical differentiation, and differentiation is an ill-posed operation on sampled data. Consider a displacement record contaminated by an additive perturbation $\delta(t) = \varepsilon \sin(\omega t)$ of small amplitude ε but high frequency ω. Its contribution to the displacement is $\mathcal{O}(\varepsilon)$ and negligible. Its contribution to the acceleration is $\ddot{\delta}(t) = -\varepsilon\omega^2 \sin(\omega t)$, of magnitude $\varepsilon\omega^2$, which for ω approaching the Nyquist limit is not negligible at all. The differentiation operator acts as a high-pass filter with gain growing as ω^2, so that the very spectral region in which a sampled record is least trustworthy is the region the estimator amplifies most.

Practical consequences follow immediately. The estimator's accuracy depends on the sampling rate in a way that is not transparent to the user. It depends on the treatment of the record endpoints, where centred difference stencils and spectral differentiation schemes alike degrade. When spectral differentiation is used — multiplying the discrete Fourier transform by $\mathrm{i}\omega$ and $-\omega^2$ — it implicitly assumes periodicity of the record, and the resulting endpoint leakage must be handled by trimming, which discards data. And critically, the estimator provides no internal mechanism for distinguishing a parameter that the data determine from one they do not. It returns a number in either case, and the classical standard errors that accompany it are computed under model-adequacy assumptions that the inverse problem itself cannot verify.

1.3 The Physics-Informed Paradigm

The physics-informed neural network reformulates the problem so that numerical differentiation of the data never occurs. Rather than estimating derivatives of the measured record, one constructs a continuous surrogate for the displacement field,

$$u(t) \approx \hat{u}_\theta(t) = \mathcal{N}(t; \theta), \tag{1.4}$$

where $\mathcal{N}$ is a neural network with parameters θ. Because $\mathcal{N}$ is a composition of smooth elementary functions, its derivatives with respect to t exist analytically to all orders and are obtained by automatic differentiation — the exact application of the chain rule through the computational graph, accurate to machine precision, with no stencil, no truncation error, and no dependence on sample spacing.

The physics enters as a constraint on that surrogate. Define the residual of the equation of motion,

$$r(t;\theta,K,C) = M\,\frac{\mathrm{d}^2\hat{u}_\theta}{\mathrm{d}t^2} + C\,\frac{\mathrm{d}\hat{u}_\theta}{\mathrm{d}t} + K\,\hat{u}_\theta(t) - F(t), \tag{1.5}$$

and require that it vanish at a set of collocation points $\{t_j\}_{j=1}^{N_f}$ distributed through the observation window. These collocation points are not data; they carry no measurement, and may be placed at arbitrary density independent of the sampling rate of the record. The network is then trained against a composite objective that simultaneously demands agreement with the observed displacements and satisfaction of the governing physics.

The step that converts this from a forward surrogate into an inverse solver is to promote K and C from fixed coefficients to trainable parameters, optimised by the same gradient descent that trains the network weights. The physics residual is the only term in which K and C appear. Consequently, the gradient signal that moves them derives entirely from the requirement that the surrogate satisfy Newton's second law — not from any fit to the displacement data, which the network can achieve at essentially any parameter values by adjusting θ alone. This separation is the conceptual heart of the method and is developed formally in Section 3.10.

Three properties recommend this formulation for structural identification. First, the derivative estimates are exact with respect to the surrogate, eliminating the differentiation-amplification pathology entirely. Second, the surrogate is a global fit, so that information from the whole record informs the estimate at every point, in contrast to the local stencils of finite differencing. Third — and this proves consequential in the results — the optimisation trajectory of each identified parameter is itself observable. A parameter that the data constrain converges and remains stable; a parameter that the data leave underdetermined continues to drift. The method thus carries an intrinsic diagnostic of identifiability that the closed-form least-squares estimator lacks.

Because this work sits at the junction of two literatures with distinct vocabularies, Figure 1.1 sets out the correspondence explicitly. Each row pairs a construct of structural dynamics with the mechanism that realises it inside the learning formulation, and names the section in which that mechanism is derived. The correspondence is not merely terminological: in every row the structural requirement dictates the algorithmic choice, rather than the reverse. The requirement that a passive structure dissipate energy is what motivates the softplus reparameterisation; the oscillatory character of the response is what compels the Fourier encoding; and the observability condition on damping is what explains, in advance of any experiment, the convergence behaviour reported in Section 5.6.

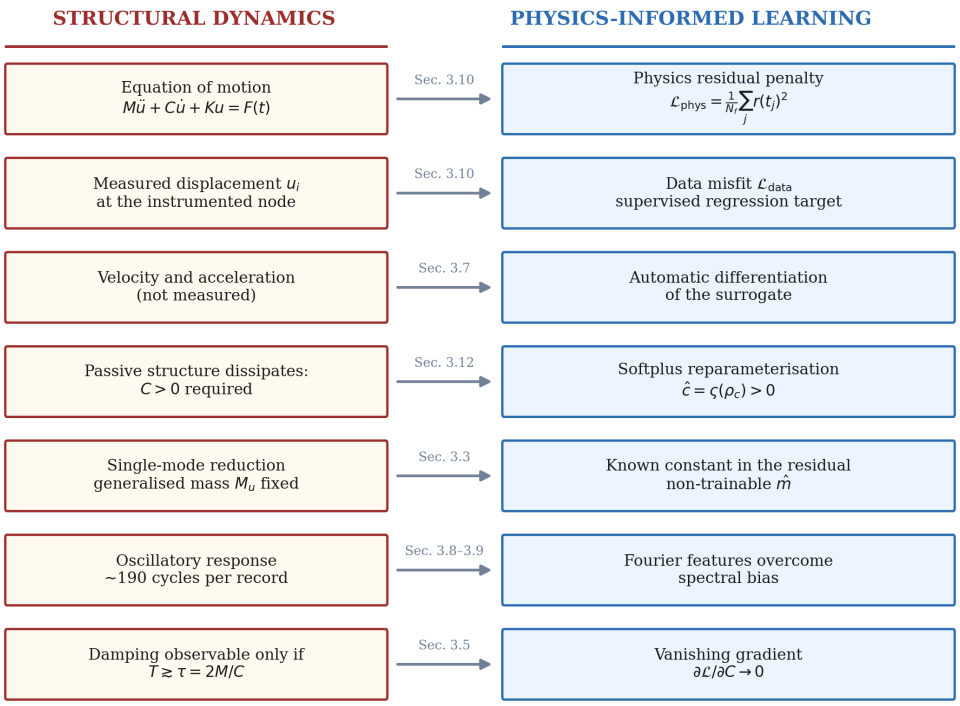


Figure 1.1. Correspondence between structural dynamics and the physics-informed learning formulation. Each structural construct on the left is realised by the mechanism on the right, with the section of derivation indicated. The mapping is directional: the physics determines the algorithmic choice in every case.

1.4 Scope, Objectives, and Contributions

The system studied is a finite element model of a steel support table of total structural mass 78,500 kg. A transient analysis applies a harmonic excitation at 15 Hz, and the displacement response is extracted at Node 3575 in each of the three coordinate directions. A separate modal analysis of the same structure provides natural frequencies and generalised masses, furnishing an independent reference against which identified stiffnesses may be compared.

The objectives of this chapter are:

1. To formulate the single-degree-of-freedom identification problem rigorously as a modal reduction of the full finite element system, making explicit the assumptions under which a single mode may represent the response at a single node, and the coordinate dependence of the resulting generalised mass.
2. To derive the complete mathematical machinery of the identification: the composite loss functional, the Fourier feature encoding required to overcome spectral bias on a record spanning approximately 190 response cycles, the non-dimensionalisation that conditions the optimisation, and the Adam update rule that drives it.
3. To demonstrate that a single direction-agnostic implementation — one code path, one architecture, one loss, switched between directions by configuration alone — identifies structural parameters in all three coordinate directions without modification.
4. To validate the identified parameters against two independent references: a classical least-squares estimator evaluated under a consistent force convention, and the modal stiffness derived from the finite element modal analysis.
5. To establish, through a systematic window-length study, the physical principle governing which parameters are identifiable from which observation records, and to show that the method's convergence diagnostics correctly reflect that principle.

The principal contributions are the direction-agnostic pipeline itself, the demonstration of accurate stiffness recovery in all three directions from displacement alone, and the identification of the decay-time-constant criterion that separates the identifiable-damping regime from the unidentifiable one.

1.5 Organisation of the Chapter

Chapter 2 reviews the relevant literature. Chapter 3 develops the theoretical foundations in full, beginning from the semi-discrete equations of motion and proceeding through modal reduction, the exact forced-damped solution, spectral bias, Fourier features, and the optimisation mechanics. Chapter 4 details the methodology and the computational implementation. Chapter 5 presents and discusses the results across all three directions. Chapter 6 concludes.

Chapter 2

Literature Review

2.1 Foundations of Physics-Informed Machine Learning

The proposal that a neural network might be trained to satisfy a differential equation rather than merely to interpolate data predates the modern deep learning era. Lagaris et al. (1998) constructed trial solutions to ordinary and partial differential equations as neural networks whose architecture enforced boundary conditions exactly, training the remaining freedom against the equation residual. The approach was limited by the optimisation technology of its time and by the shallow architectures then practical, but the formulation — residual minimisation over a parameterised function class — is recognisably that used today.

The modern formulation is due to Raissi et al. (2019), who introduced the term *physics-informed neural network* and, critically, framed both the forward and the inverse problem within a single architecture. Their central observation was that automatic differentiation, developed for backpropagation through network weights, applies equally to differentiation with respect to network inputs. A network $\hat{u}_\theta(t)$ therefore yields $\mathrm{d}\hat{u}_\theta/\mathrm{d}t$, $\mathrm{d}^2\hat{u}_\theta/\mathrm{d}t^2$, and any other required derivative at machine precision and negligible marginal cost. The physics residual becomes a computable, differentiable function of θ, and the whole problem reduces to unconstrained optimisation.

The inverse formulation in that work is the direct antecedent of the present study. By treating an unknown coefficient of the governing equation as a trainable parameter, Raissi et al. demonstrated recovery of the Reynolds number in Navier–Stokes flow and of coefficients in the Korteweg–de Vries equation from sparse observations. The structural identification problem addressed here is of the same type, with K and C playing the role of the unknown coefficients.

Karniadakis et al. (2021) survey the field's subsequent expansion and articulate the governing principle: physics-informed learning injects inductive bias into an otherwise unconstrained function approximator, so that the hypothesis space is narrowed to functions consistent with known conservation laws. In the data-sparse regime characteristic of structural monitoring, where the number of instrumented degrees of freedom is minute relative to the model's dimension, this bias is not a convenience but a necessity.

2.2 Inverse Parameter Identification with PINNs

Applications of PINNs to structural and mechanical parameter identification have developed rapidly. Zhang et al. (2020) applied physics-informed multi-LSTM networks to the metamodelling of nonlinear structures under seismic excitation, establishing the viability of physics-constrained learning for civil structures. Haghighat et al. (2021) developed a PINN framework

for solid mechanics that identifies elastic constants from displacement fields, demonstrating that the method extends naturally from scalar coefficients to constitutive parameters.

Several authors have examined the specific question of damping identification. The consensus in this literature — arrived at independently of the present work and consistent with its findings — is that damping is substantially harder to identify than stiffness. Stiffness governs the oscillation frequency and thus manifests in every cycle of the response; damping governs the rate of amplitude decay and manifests only over the timescale on which that decay is appreciable. Where the observation window is short relative to the decay time constant, the damping term contributes negligibly to the residual and the corresponding gradient is correspondingly weak. This is a property of the information content of the measurement, not of the estimator, and it applies with equal force to classical estimators — a point developed quantitatively in Section 3.5.

2.3 Spectral Bias and the Fourier Feature Remedy

A finding of central practical importance to this work concerns the frequency content that a standard neural network can represent. Rahaman et al. (2019) demonstrated empirically that deep networks trained by gradient descent learn low-frequency components of a target function substantially before high-frequency ones, terming this *spectral bias*. Basri et al. (2019) and Cao et al. (2019) established the theoretical basis through neural tangent kernel (NTK) analysis: the eigenvalues of the NTK associated with high-frequency eigenfunctions decay rapidly, and since convergence along each eigendirection proceeds at a rate proportional to the corresponding eigenvalue, high-frequency content is learned orders of magnitude more slowly.

The consequence for oscillatory structural response is severe. A record spanning 10 s contains 150 cycles of the 15 Hz drive and, at the ≈19 Hz frequency of the structural response, approximately 190 oscillations in total. A standard fully connected network with coordinate input t must represent a function oscillating some 190 times across its domain, and spectral bias renders this impractical within any reasonable iteration budget. The failure mode observed in preliminary runs of the present study — successful representation up to approximately 30 oscillations, breakdown beyond — matches this prediction.

Tancik et al. (2020) provided the remedy, building on the random Fourier feature construction of Rahimi and Recht (2008). By passing the input coordinate through a sinusoidal encoding,

$$\gamma(t) = \left[\sin(2\pi \mathbf{b}\, t),\ \cos(2\pi \mathbf{b}\, t)\right]^{\mathsf{T}}, \tag{2.1}$$

with $\mathbf{b}$ a vector of frequencies, the effective NTK is transformed into a stationary kernel whose bandwidth is controlled by the spectrum of $\mathbf{b}$. High-frequency content becomes learnable at rates comparable to low-frequency content. Wang et al. (2021) applied this construction specifically to PINNs, confirming that Fourier feature embeddings resolve the multi-scale failure modes that otherwise afflict physics-informed training. The present work adopts this encoding with a frequency band chosen to bracket the structure's response spectrum.

2.4 The DeepXDE Framework

Lu et al. (2021) introduced DeepXDE, a library that abstracts the construction of physics-informed problems: geometry specification, residual definition, boundary and initial conditions, collocation point sampling, and the training loop. Its support for inverse problems through trainable variables makes it directly suited to identification, and it is the implementation substrate of this work.

One characteristic of the framework requires explicit methodological attention. DeepXDE's default reporting evaluates a "test" loss on points drawn from the same distribution as the training collocation set, which does not constitute held-out validation in the statistical sense. The present methodology therefore constructs an explicit holdout partition, withholding a deterministic subset of observed samples from the data loss entirely and reporting metrics on those samples separately. This is detailed in Section 4.7.

2.5 Classical System Identification: The Comparison Baseline

The classical literature provides the standard against which the PINN must be measured. Ljung (1999) is the canonical treatment of system identification generally; Ewins (2000) and Maia and Silva (1997) develop modal testing and experimental modal analysis specifically for structures. The subspace identification methods of Van Overschee and De Moor (1996) and the stochastic subspace identification of Peeters and De Roeck (1999) represent the mature state of output-only identification for civil structures.

For the single-degree-of-freedom case considered here, the appropriate baseline is the direct least-squares estimator described in Section 1.2, with derivatives obtained by spectral differentiation. This estimator is used throughout as an independent check. Its known sensitivities — to the force convention assumed, to endpoint leakage, and to the adequacy of the single-degree-of-freedom reduction — are treated explicitly rather than assumed away, and the comparison is conducted under a matched force convention so that the two estimators address the same problem.

2.6 Multi-Degree-of-Freedom Systems and Modal Reduction

The reduction of a multi-degree-of-freedom finite element model to a single-degree-of-freedom oscillator is classical, and its assumptions are well documented by Chopra (2017) and Clough and Penzien (1993). The essential requirement is modal separation: the reduction is valid to the extent that the response in the frequency band of interest is dominated by a single mode. Where modes are closely spaced, or where the excitation drives several modes comparably, a single-mode reduction misrepresents the dynamics regardless of how well any individual parameter is fitted.

The generalised quantities arising from this reduction — generalised mass, generalised stiffness, generalised force — are coordinate-dependent. They depend on the normalisation of the mode

shape and on the particular physical degree of freedom at which the response is observed. This dependence is a frequent source of confusion in the applied literature and is derived explicitly in Section 3.3, because it bears directly on the interpretation of the widely differing effective masses encountered across the three coordinate directions in this study.

2.7 Identified Research Gap

Three observations motivate the present contribution. First, the PINN identification literature reports predominantly single-direction or single-parameter studies; systematic evaluation of one unmodified pipeline across multiple spatial directions of the same structure is not established. Second, the interaction between spectral bias and long oscillatory records — the regime in which structural monitoring data actually arrive — is acknowledged but seldom studied quantitatively against observation window length. Third, and most consequentially, the literature reports identification successes but rarely characterises the boundary of identifiability: the conditions under which a given parameter can be recovered from a given record at all. This chapter addresses all three.

Chapter 3

Theoretical Foundations

3.1 From the Continuum to the Semi-Discrete Equations of Motion

The starting point is the principle of virtual work for a continuous elastic body Ω undergoing small-displacement motion. For an admissible virtual displacement field $\delta\mathbf{u}$,

$$\int_\Omega \rho\,\delta\mathbf{u}\cdot\ddot{\mathbf{u}}\,\mathrm{d}\Omega + \int_\Omega \delta\boldsymbol{\varepsilon} : \boldsymbol{\sigma}\,\mathrm{d}\Omega + \int_\Omega \eta\,\delta\mathbf{u}\cdot\dot{\mathbf{u}}\,\mathrm{d}\Omega = \int_\Omega \delta\mathbf{u}\cdot\mathbf{f}_b\,\mathrm{d}\Omega + \int_{\Gamma_N} \delta\mathbf{u}\cdot\bar{\mathbf{t}}\,\mathrm{d}\Gamma, \tag{3.1}$$

where ρ is mass density, $\boldsymbol{\sigma}$ the Cauchy stress, $\boldsymbol{\varepsilon}$ the infinitesimal strain, η a distributed viscous dissipation coefficient, $\mathbf{f}_b$ the body force, and $\bar{\mathbf{t}}$ the surface traction on the Neumann boundary Γ_N.

Introducing a finite element discretisation $\mathbf{u}(\mathbf{x},t) \approx \mathbf{N}(\mathbf{x})\,\mathbf{q}(t)$, with $\mathbf{N}$ the shape function matrix and $\mathbf{q}$ the vector of nodal degrees of freedom, and applying the strain–displacement relation $\boldsymbol{\varepsilon} = \mathbf{B}\mathbf{q}$ together with the linear elastic constitutive law $\boldsymbol{\sigma} = \mathbf{D}\boldsymbol{\varepsilon}$, the arbitrariness of $\delta\mathbf{q}$ yields the semi-discrete system

$$\mathbf{M}\ddot{\mathbf{q}}(t) + \mathbf{C}\dot{\mathbf{q}}(t) + \mathbf{K}\mathbf{q}(t) = \mathbf{f}(t), \tag{3.2}$$

with

$$\mathbf{M} = \int_\Omega \rho\,\mathbf{N}^{\mathsf{T}}\mathbf{N}\,\mathrm{d}\Omega, \qquad \mathbf{C} = \int_\Omega \eta\,\mathbf{N}^{\mathsf{T}}\mathbf{N}\,\mathrm{d}\Omega, \qquad \mathbf{K} = \int_\Omega \mathbf{B}^{\mathsf{T}}\mathbf{D}\,\mathbf{B}\,\mathrm{d}\Omega, \tag{3.3}$$

and $\mathbf{f}(t)$ the consistent nodal force vector. The matrices $\mathbf{M}$ and $\mathbf{K}$ are symmetric and positive definite for a properly restrained structure. This is the system that the finite element transient analysis integrates, and it is the system from which the observed single-node response is extracted.

3.2 Modal Decomposition and Single-Mode Reduction

The identification problem is posed on a scalar oscillator, not on the n-dimensional system above. The bridge between them is modal decomposition, and its assumptions must be made explicit because they govern the interpretation of every identified quantity.

Consider the undamped free vibration problem obtained by setting $\mathbf{C} = \mathbf{0}$ and $\mathbf{f} = \mathbf{0}$. Seeking solutions of the form $\mathbf{q}(t) = \boldsymbol{\phi}\,e^{\mathrm{i}\omega t}$ yields the generalised eigenvalue problem

$$(\mathbf{K} - \omega^2\mathbf{M})\boldsymbol{\phi} = \mathbf{0}. \tag{3.4}$$

This admits n eigenpairs $(\omega_r^2, \boldsymbol{\phi}_r)$, $r = 1,\dots,n$, with real non-negative eigenvalues by the symmetry and positive definiteness of $\mathbf{M}$ and $\mathbf{K}$. The eigenvectors satisfy the orthogonality

relations

$$\boldsymbol{\phi}_s^{\mathsf{T}}\mathbf{M}\boldsymbol{\phi}_r = 0, \qquad \boldsymbol{\phi}_s^{\mathsf{T}}\mathbf{K}\boldsymbol{\phi}_r = 0, \qquad r \neq s, \tag{3.5}$$

which follow directly from the eigenvalue problem: premultiplying the r-th equation by $\boldsymbol{\phi}_s^{\mathsf{T}}$ and the s-th by $\boldsymbol{\phi}_r^{\mathsf{T}}$, transposing the latter, and subtracting gives $(\omega_r^2 - \omega_s^2)\,\boldsymbol{\phi}_s^{\mathsf{T}}\mathbf{M}\boldsymbol{\phi}_r = 0$, from which the $\mathbf{M}$-orthogonality follows for distinct eigenvalues and the $\mathbf{K}$-orthogonality then follows immediately.

Define the modal (generalised) mass and stiffness

$$M_r = \boldsymbol{\phi}_r^{\mathsf{T}}\mathbf{M}\boldsymbol{\phi}_r, \qquad K_r = \boldsymbol{\phi}_r^{\mathsf{T}}\mathbf{K}\boldsymbol{\phi}_r = \omega_r^2 M_r. \tag{3.6}$$

Expanding the response in the modal basis, $\mathbf{q}(t) = \sum_s \boldsymbol{\phi}_s\,\eta_s(t)$, substituting into Equation (3.2), and premultiplying by $\boldsymbol{\phi}_r^{\mathsf{T}}$ gives

$$M_r\ddot{\eta}_r + \sum_s (\boldsymbol{\phi}_r^{\mathsf{T}}\mathbf{C}\boldsymbol{\phi}_s)\dot{\eta}_s + K_r\eta_r = \boldsymbol{\phi}_r^{\mathsf{T}}\mathbf{f}(t). \tag{3.7}$$

Under the classical damping assumption — that $\mathbf{C}$ is diagonalised by the same transformation, as holds exactly for Rayleigh damping $\mathbf{C} = \alpha\mathbf{M} + \beta\mathbf{K}$ — the damping coupling terms vanish for $s \neq r$, and the system decouples completely into n independent scalar oscillators

$$M_r\ddot{\eta}_r + C_r\dot{\eta}_r + K_r\eta_r = F_r(t), \qquad C_r = \boldsymbol{\phi}_r^{\mathsf{T}}\mathbf{C}\boldsymbol{\phi}_r, \quad F_r(t) = \boldsymbol{\phi}_r^{\mathsf{T}}\mathbf{f}(t). \tag{3.8}$$

The single-degree-of-freedom reduction retains one such equation. Its validity rests on *modal dominance*: that within the frequency band of the excitation and the measurement, one mode r contributes the overwhelming majority of the response, so that $\mathbf{q}(t) \approx \boldsymbol{\phi}_r\,\eta_r(t)$. The extent to which this holds is an empirical property of the structure and the loading, assessable from the response spectrum, and it is examined directly for each direction in Chapter 5.

3.3 The Generalised Force and Physical-Coordinate Recovery

The identification operates not on the modal coordinate η_r, which is unmeasurable, but on the physical displacement at the observation node. Let j index the observed degree of freedom and let $\phi_{r,j}$ denote the j-th component of $\boldsymbol{\phi}_r$. Under single-mode dominance,

$$u(t) = q_j(t) \approx \phi_{r,j}\,\eta_r(t). \tag{3.9}$$

Substituting $\eta_r = u/\phi_{r,j}$ into Equation (3.8) and dividing through by $\phi_{r,j}$ recasts the oscillator in terms of the measured coordinate:

$$M_u\ddot{u} + C_u\dot{u} + K_u u = F_u(t), \qquad M_u = \frac{M_r}{\phi_{r,j}^2}, \quad C_u = \frac{C_r}{\phi_{r,j}^2}, \quad K_u = \frac{K_r}{\phi_{r,j}^2}, \quad F_u(t) = \frac{\boldsymbol{\phi}_r^{\mathsf{T}}\mathbf{f}(t)}{\phi_{r,j}}. \tag{3.10}$$

This is the equation that the identification actually addresses, and three consequences follow that are essential to interpreting the results.

First, *the effective mass is coordinate-dependent.* The quantity M_u depends on both the mode-shape normalisation and the amplitude of that mode shape at the observation node. If the modal vectors are mass-normalised so that $M_r = 1$, then $M_u = 1/\phi_{r,j}^2$ exactly. A mode with small participation at the observation node — a node near a modal nodal point — yields a large M_u. These generalised masses are therefore not total structural masses, nor effective modal mass percentages, and they may legitimately exceed the total mass of the structure by orders of magnitude. This explains the wide disparity in the effective masses across the three directions in this study without any inconsistency: it is the expected behaviour of the transformation.

Second, *the identified stiffness inherits the same scaling*, $K_u = \omega_r^2 M_u$, so that the ratio $K_u/M_u = \omega_r^2$ is invariant and the identified natural frequency is the physically meaningful comparison quantity across directions.

Third, *the generalised force F_u is in general not the applied physical force.* It is the projection of the full spatial load vector onto the mode shape, rescaled by the observation-node component. For a load applied in one direction and a mode responding in another, the generalised force is nonzero — cross-axis modal forcing is generic, not exceptional — but its magnitude and sign depend on the spatial load distribution, the mode shape, and the normalisation convention. Establishing F_u therefore requires the spatial force vector and the mode shape, not merely the scalar amplitude of the applied load.

3.4 Exact Solution of the Forced Damped Oscillator

The reduced equation admits a closed-form solution under harmonic excitation, which is required both for the physical interpretation of the identified parameters and for the verification calculations of Chapter 5. Consider

$$M\ddot{u} + C\dot{u} + Ku = F_0 \sin(\Omega t). \tag{3.11}$$

Define the undamped natural frequency and damping ratio

$$\omega_n = \sqrt{\frac{K}{M}}, \qquad \zeta = \frac{C}{2\sqrt{KM}}, \tag{3.12}$$

so that the equation may be written in standard form $\ddot{u} + 2\zeta\omega_n\dot{u} + \omega_n^2 u = (F_0/M)\sin(\Omega t)$.

Particular (steady-state) solution. Seeking $u_p(t) = a\cos(\Omega t) + b\sin(\Omega t)$ and substituting, the coefficients of $\sin(\Omega t)$ and $\cos(\Omega t)$ must balance independently. Writing $K_d = K - M\Omega^2$ for the dynamic stiffness and collecting terms yields

$$\begin{bmatrix} K_d & C\Omega \\ -C\Omega & K_d \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ F_0 \end{bmatrix}, \tag{3.13}$$

with solution

$$a = \frac{-F_0\, C\Omega}{K_d^2 + (C\Omega)^2}, \qquad b = \frac{F_0\, K_d}{K_d^2 + (C\Omega)^2}. \tag{3.14}$$

Writing $\rho_\Omega = \Omega/\omega_n$ for the frequency ratio, the steady-state response is $u_p(t) = U\sin(\Omega t - \varphi)$ with amplitude and phase

$$U = \frac{F_0/K}{\sqrt{\left(1-\rho_\Omega^2\right)^2 + \left(2\zeta\rho_\Omega\right)^2}}, \qquad \tan\varphi = \frac{2\zeta\rho_\Omega}{1-\rho_\Omega^2}. \tag{3.15}$$

Homogeneous (transient) solution. The complementary solution satisfies $M\ddot{u}+C\dot{u}+Ku = 0$, with characteristic roots

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}. \tag{3.16}$$

For the underdamped case $\zeta < 1$, which holds for all structural systems of interest, $s_{1,2} = -\zeta\omega_n \pm \mathrm{i}\,\omega_d$ with $\omega_d = \omega_n\sqrt{1-\zeta^2}$, and

$$u_h(t) = e^{-\zeta\omega_n t}\big[A\cos(\omega_d t) + B\sin(\omega_d t)\big]. \tag{3.17}$$

Complete solution. The full response is $u(t) = u_h(t) + u_p(t)$, with A and B fixed by the initial conditions $u(0)$ and $\dot{u}(0)$. This expression is used directly in Section 5.9 as an exact-dynamics consistency check on the identified parameters.

3.5 The Decay Envelope and Parameter Observability

The structure of the complete solution yields the central physical result of this chapter, and it is worth deriving explicitly because it governs the interpretation of every damping estimate reported later.

The transient component is modulated by the envelope $e^{-\zeta\omega_n t}$, with $\zeta\omega_n = C/(2M)$. Define the decay time constant

$$\tau = \frac{1}{\zeta\omega_n} = \frac{2M}{C}, \tag{3.18}$$

the time over which the free-response amplitude falls by a factor of e. Over an observation window of length T, the total fractional amplitude decay is

$$\Delta = 1 - e^{-T/\tau}, \tag{3.19}$$

and for $T \ll \tau$ this admits the expansion

$$\Delta = \frac{T}{\tau} - \frac{1}{2}\Big(\frac{T}{\tau}\Big)^2 + \mathcal{O}\Big((T/\tau)^3\Big) \approx \frac{T}{\tau} = \frac{CT}{2M}. \tag{3.20}$$

The damping coefficient enters the observable response, at leading order, only through the dimensionless ratio T/τ. Two regimes follow.

When $T/\tau \gtrsim 1$, the decay is substantial, the envelope is clearly resolved within the record, and

C is well determined. The residual is materially sensitive to perturbations in C, so the gradient $\partial\mathcal{L}/\partial C$ is large and the optimisation converges tightly.

When $T/\tau \ll 1$, the fractional decay across the entire record is $\Delta \approx T/\tau$ and may lie below the level at which it can be distinguished from other modelling discrepancies. The residual becomes nearly independent of C; the gradient correspondingly vanishes; and the optimisation has no restoring force toward any particular value. The parameter is then said to be *weakly identifiable* or *unidentifiable* from that record.

It is essential to recognise that this is a property of the information content of the measurement, not a deficiency of any particular estimator. A classical least-squares estimator applied in the $T/\tau \ll 1$ regime does not fail gracefully — it returns a number, and that number may be negative, since nothing in the unconstrained normal equations enforces $C > 0$. A negative damping estimate is a diagnostic signature of this regime. It is not a physical property of a passive structure, which necessarily dissipates energy and therefore has $C > 0$; it is the estimator reporting that the data do not determine the parameter.

Stiffness stands in complete contrast. It enters through $\omega_n = \sqrt{K/M}$, which sets the oscillation frequency itself. A perturbation δK produces a frequency shift $\delta\omega_n = \omega_n\,\delta K/(2K)$ and hence a phase drift accumulating as

$$\delta\varphi(t) = \delta\omega_n\, t = \frac{\omega_n t}{2}\,\frac{\delta K}{K}, \tag{3.21}$$

which becomes detectable within a handful of cycles. Stiffness is therefore identifiable from short records, and this asymmetry between the two parameters is the governing principle behind the window-length study reported in Section 5.7.

3.6 Energy Balance of the Identified System

A complementary view of the same asymmetry follows from energy accounting. Multiplying the equation of motion by $\dot{u}$ and integrating over $[0, T]$,

$$\int_0^T M\ddot{u}\,\dot{u}\,\mathrm{d}t + \int_0^T C\dot{u}^2\,\mathrm{d}t + \int_0^T Ku\,\dot{u}\,\mathrm{d}t = \int_0^T F\,\dot{u}\,\mathrm{d}t. \tag{3.22}$$

The first and third integrands are exact derivatives, giving

$$\Big[\tfrac{1}{2}M\dot{u}^2\Big]_0^T + \underbrace{\int_0^T C\dot{u}^2\,\mathrm{d}t}_{E_D(T)} + \Big[\tfrac{1}{2}Ku^2\Big]_0^T = \int_0^T F\,\dot{u}\,\mathrm{d}t. \tag{3.23}$$

The dissipation term E_D is the only one that is not a boundary term: kinetic and potential energy are recovered at each cycle, whereas dissipated energy accumulates monotonically. Over a single cycle of steady-state harmonic response at frequency Ω and amplitude U, the dissipated energy is

$$E_D^{\text{cycle}} = \int_0^{2\pi/\Omega} C\dot{u}^2\,\mathrm{d}t = \pi C\Omega U^2, \tag{3.24}$$

while the peak stored elastic energy is $V_{\max} = \frac{1}{2}KU^2$. Their ratio,

$$\frac{E_D^{\text{cycle}}}{V_{\max}} = \frac{2\pi C\Omega}{K} \xrightarrow{\Omega=\omega_n} 4\pi\zeta, \tag{3.25}$$

shows that for lightly damped structures, with ζ of order 10^{-3}, the energy dissipated per cycle is a fraction of order $4\pi\zeta$ — about 3 % at $\zeta = 0.25\,\%$ — of the energy stored. The damping term is a small perturbation on the cycle-by-cycle energy budget, and its effect becomes measurable only through accumulation over many cycles. This is the energetic statement of the same observability principle derived kinematically in Section 3.5, and it makes explicit why a record must be long relative to τ for C to be determined.

3.7 Neural Network Function Approximation

The displacement surrogate is a fully connected feedforward network. Let L denote the number of layers, $\mathbf{W}^{(\ell)}$ and $\mathbf{b}^{(\ell)}$ the weights and biases of layer ℓ, and σ a nonlinear activation applied componentwise. With $\mathbf{z}^{(0)} = \gamma(t)$ the encoded input,

$$\mathbf{z}^{(\ell)} = \sigma\big(\mathbf{W}^{(\ell)}\mathbf{z}^{(\ell-1)} + \mathbf{b}^{(\ell)}\big), \quad \ell = 1, \dots, L-1, \qquad \hat{u}_\theta(t) = \mathbf{W}^{(L)}\mathbf{z}^{(L-1)} + \mathbf{b}^{(L)}, \tag{3.26}$$

with $\theta = \{\mathbf{W}^{(\ell)}, \mathbf{b}^{(\ell)}\}_{\ell=1}^{L}$.

The universal approximation theorem (Cybenko, 1989; Hornik, 1991) guarantees that such a network with a single sufficiently wide hidden layer and a non-polynomial activation approximates any continuous function on a compact domain to arbitrary accuracy. For the physics-informed setting a stronger requirement applies: the network must be differentiable to the order appearing in the governing equation, here second order, and the derivatives themselves must be well approximated. This motivates smooth activations. The present work uses $\sigma = \sin$, following Sitzmann et al. (2020), whose sinusoidal representation networks (SIRENs) were shown to represent signals and their derivatives with markedly higher fidelity than piecewise-linear alternatives. The choice is doubly appropriate here: the derivative of a sinusoid is a sinusoid, so differentiation does not degrade the representational class, and the target function is itself oscillatory.

Automatic differentiation supplies the required derivatives exactly. Writing the network as a composition $\hat{u}_\theta = f_L \circ f_{L-1} \circ \cdots \circ f_1 \circ \gamma$, the first derivative follows by the chain rule,

$$\frac{\mathrm{d}\hat{u}_\theta}{\mathrm{d}t} = \frac{\partial f_L}{\partial \mathbf{z}^{(L-1)}} \frac{\partial f_{L-1}}{\partial \mathbf{z}^{(L-2)}} \cdots \frac{\partial f_1}{\partial \mathbf{z}^{(0)}} \frac{\mathrm{d}\gamma}{\mathrm{d}t}, \tag{3.27}$$

and the second by applying the same construction to the computational graph of the first. Each derivative is exact to floating-point precision. No stencil, no truncation error, and no dependence on the sampling interval of the measured record enters at any point — which is precisely the pathology of classical estimators that this formulation is designed to eliminate.

3.8 Spectral Bias: A Neural Tangent Kernel Derivation

The necessity of the Fourier encoding is not heuristic; it follows from the training dynamics of wide networks, and the derivation is given here because the encoding's design parameters are chosen directly from it.

Consider gradient descent on a loss $\mathcal{L}(\theta) = \frac{1}{2}\sum_i \big(\hat{u}_\theta(t_i) - u_i\big)^2$ with learning rate η. In the infinite-width limit, Jacot et al. (2018) showed that the network's evolution is governed by the neural tangent kernel

$$\Theta(t, t') = \nabla_\theta \hat{u}_\theta(t)^\mathsf{T} \nabla_\theta \hat{u}_\theta(t'), \tag{3.28}$$

which remains approximately constant throughout training. The function-space dynamics then become linear:

$$\frac{\partial \hat{u}_\theta(t)}{\partial s} = -\eta \sum_i \Theta(t, t_i)\big(\hat{u}_\theta(t_i) - u_i\big), \tag{3.29}$$

where s denotes training time. Writing $\mathbf{e}(s)$ for the vector of residuals at the training points and $\boldsymbol{\Theta}$ for the kernel Gram matrix, this is

$$\frac{\mathrm{d}\mathbf{e}}{\mathrm{d}s} = -\eta\, \boldsymbol{\Theta}\, \mathbf{e}. \tag{3.30}$$

Diagonalising $\boldsymbol{\Theta} = \sum_k \lambda_k \mathbf{v}_k \mathbf{v}_k^\mathsf{T}$ with eigenvalues λ_k and eigenvectors $\mathbf{v}_k$, the residual decomposes as

$$\mathbf{e}(s) = \sum_k e^{-\eta \lambda_k s} \big(\mathbf{v}_k^\mathsf{T} \mathbf{e}(0)\big) \mathbf{v}_k. \tag{3.31}$$

Each error mode decays at a rate proportional to its kernel eigenvalue. For a standard fully connected network with raw coordinate input, the NTK eigenfunctions are approximately sinusoids and the eigenvalues decay rapidly with frequency — for common architectures, as an inverse power law $\lambda_k \propto k^{-p}$ in the frequency index. Convergence along a high-frequency eigendirection is therefore slower by the corresponding factor, and the number of iterations required to resolve a component at frequency k grows sharply with k. This is spectral bias.

Applied to the present problem: the response record spans $T = 10\,\mathrm{s}$. At the 15 Hz drive frequency this is 150 cycles, and at the ≈19 Hz response frequency approximately 190 full oscillations. Representing this with a raw-coordinate network demands convergence at an eigendirection far into the rapidly decaying tail of the NTK spectrum. The iteration budget required is prohibitive, and the practical failure is not graceful degradation but the optimiser settling into a low-frequency solution that ignores the oscillation entirely.

3.9 Fourier Feature Encoding

The remedy of Tancik et al. (2020) is to modify the kernel directly by preprocessing the input. Define the encoding

$$\gamma(t) = \big[\sin(2\pi b_1 t), \cos(2\pi b_1 t), \ldots, \sin(2\pi b_m t), \cos(2\pi b_m t)\big]^\mathsf{T} \in \mathbb{R}^{2m}, \tag{3.32}$$

with frequencies b_i drawn from a chosen band. The essential property is revealed by computing the inner product of two encoded inputs:

$$\gamma(t)^{\mathsf{T}}\gamma(t') = \sum_{i=1}^{m}\left[\sin(2\pi b_i t)\sin(2\pi b_i t') + \cos(2\pi b_i t)\cos(2\pi b_i t')\right] = \sum_{i=1}^{m}\cos\big(2\pi b_i\,(t-t')\big), \quad (3.33)$$

by the cosine difference identity. The result depends on t and t' only through their difference $t - t'$: the induced kernel is *stationary* (shift-invariant). This is the random Fourier feature construction of Rahimi and Recht (2008), and in the limit of many frequencies drawn from a distribution $p(b)$, Bochner's theorem identifies the limiting kernel as the Fourier transform of p:

$$k(\Delta t) = \int p(b)\cos(2\pi b\,\Delta t)\,\mathrm{d}b. \quad (3.34)$$

The composed kernel of the encoded network is therefore no longer the rapidly decaying NTK of the raw-coordinate network but one whose bandwidth is set by the support of $p(b)$. Frequencies within that band acquire comparable eigenvalues and are learned at comparable rates. Spectral bias is not eliminated in principle — it is relocated, so that the band of interest sits where the kernel spectrum is flat.

This makes the design rule explicit: *the encoding band must bracket the spectral content of the target response.* Too narrow a band omits response content; too wide a band admits high-frequency directions that serve only to fit noise and degrade conditioning. For the X and Y directions the encoding uses $m = 128$ frequency pairs spanning 5–40 Hz, chosen to bracket the 15 Hz drive frequency together with the structural response content observed between approximately 18 and 20 Hz (Figure 3.1), with margin at both ends. The Z direction requires a wider band for the reason given in Section 4.3.

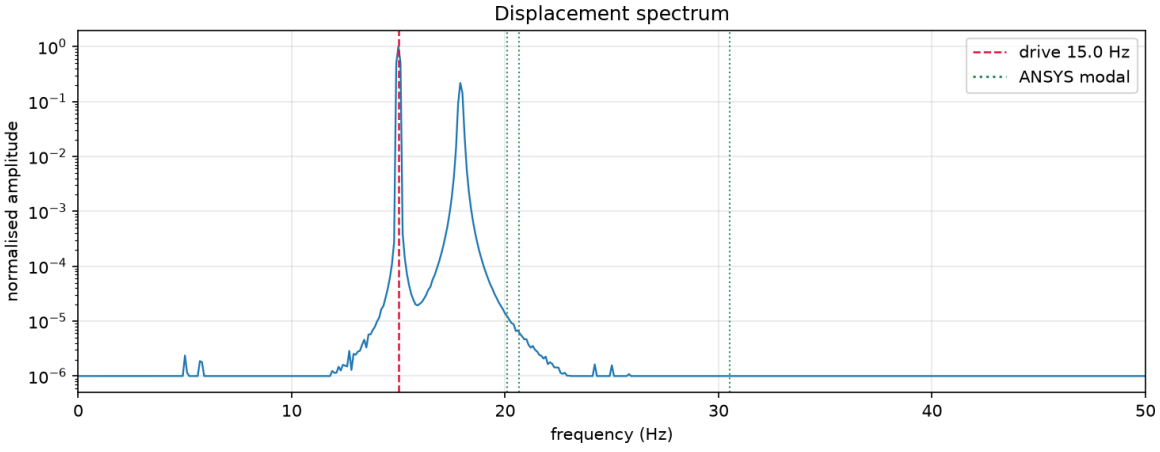


Figure 3.1. Response spectrum for the X direction, showing the 15 Hz drive component, the structural response content near 17.9 Hz, and the modal reference lines. The Fourier encoding band is selected to bracket this content.

The complete forward architecture, assembled from the components derived in Sections 3.7 to 3.9, is shown in Figure 3.3. Three features of the construction warrant emphasis. The encoding lifts the scalar time coordinate into $\mathbb{R}^{256}$ before it reaches any trainable layer, so

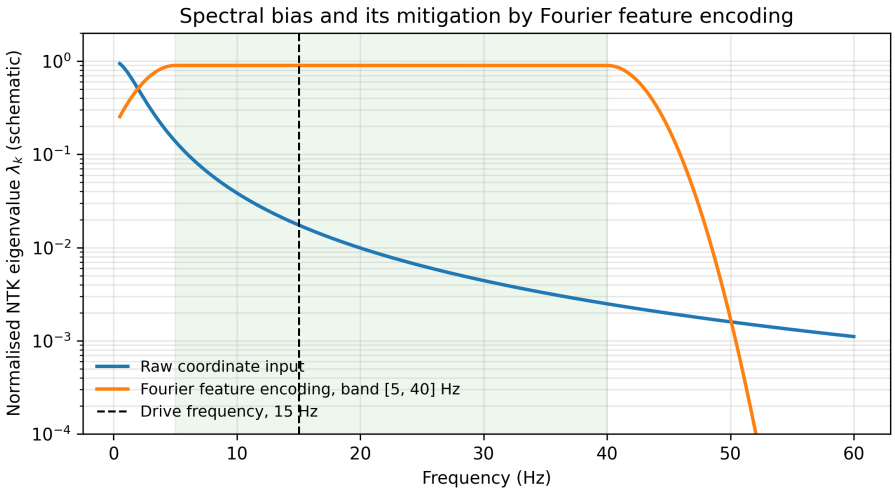


Figure 3.2. Schematic comparison of NTK eigenvalue spectra. The raw-coordinate network's eigenvalues decay rapidly with frequency, so that convergence at 15 Hz is slow. The Fourier-encoded network maintains comparable eigenvalues across the encoding band. The curves are illustrative, not computed from the trained network.

the network is never asked to represent the oscillation from a raw coordinate. The activation is sinusoidal throughout, so differentiation preserves the representational class rather than degrading it. And the identified parameters $\hat{k}$ and $\hat{c}$ are not outputs of the network: they are free scalars of the optimisation, entering only through the residual, which is the structural fact underlying the gradient analysis of Section 3.10.

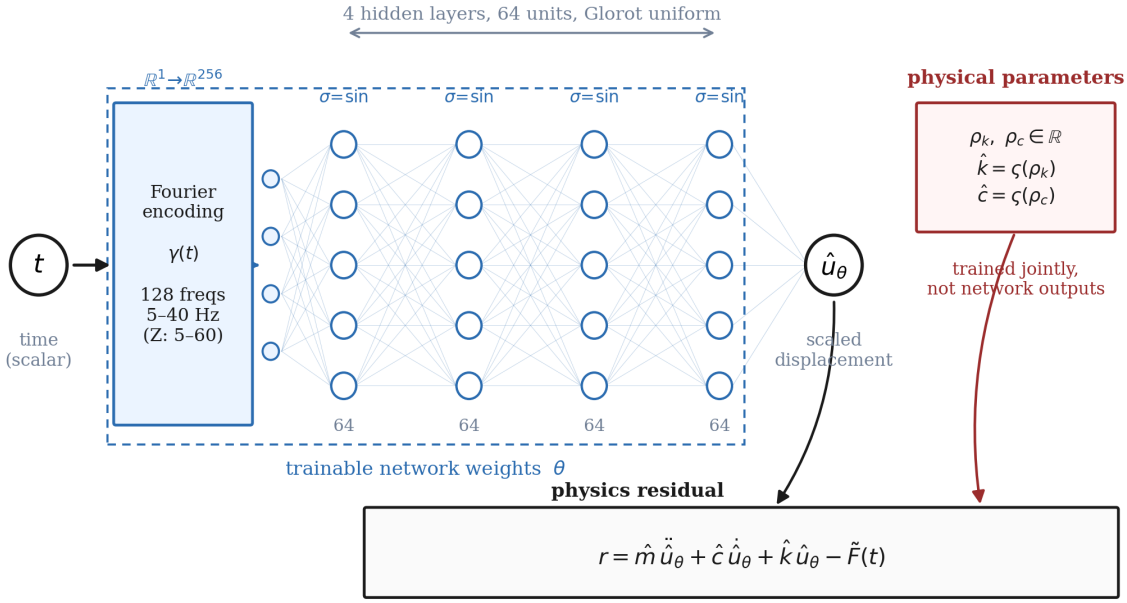


Figure 3.3. Forward architecture of the identification network. The scalar time coordinate is lifted by the Fourier encoding into a 256-dimensional representation, propagated through four fully connected layers of 64 sinusoidally activated units under Glorot uniform initialisation, and mapped to the scaled displacement $\hat{u}_\theta$. The trainable network weights are enclosed by the dashed boundary. The identified stiffness and damping enter as unconstrained latent scalars ρ_k, ρ_c transformed by the softplus map of Section 3.12; they lie outside the network and are not among its outputs, reaching the objective solely through the physics residual.

3.10 The Composite Physics-Informed Loss Functional

The identification is posed as the minimisation of a composite objective over the network parameters and the unknown physical parameters jointly.

Data term. Let $\{(t_i, u_i)\}_{i=1}^{N_d}$ denote the observed displacement samples retained for training. The data misfit is

$$\mathcal{L}_{\text{data}}(\theta) = \frac{1}{N_d} \sum_{i=1}^{N_d} \big(\hat{u}_\theta(t_i) - u_i\big)^2. \tag{3.35}$$

Physics term. Let $\{t_j\}_{j=1}^{N_f}$ denote collocation points distributed through the observation window. These are not measurements and carry no observed value; they are locations at which the governing equation is required to hold. Define the residual

$$r(t_j; \theta, K, C) = M\, \ddot{\hat{u}}_\theta(t_j) + C\, \dot{\hat{u}}_\theta(t_j) + K\, \hat{u}_\theta(t_j) - F(t_j) \tag{3.36}$$

and the physics misfit

$$\mathcal{L}_{\text{phys}}(\theta, K, C) = \frac{1}{N_f} \sum_{j=1}^{N_f} r(t_j; \theta, K, C)^2. \tag{3.37}$$

Composite objective. With weights $w_d > 0$ and $w_f > 0$,

$$\mathcal{L}(\theta, K, C) = w_d\, \mathcal{L}_{\text{data}}(\theta) + w_f\, \mathcal{L}_{\text{phys}}(\theta, K, C), \tag{3.38}$$

and the identification problem is

$$(\theta^*, K^*, C^*) = \arg\min_{\theta, K, C}\ \mathcal{L}(\theta, K, C). \tag{3.39}$$

The gradient structure is the key to the method, and deserves explicit statement. Observe that K and C appear in $\mathcal{L}_{\text{phys}}$ alone; they are entirely absent from $\mathcal{L}_{\text{data}}$. Differentiating,

$$\frac{\partial \mathcal{L}}{\partial K} = \frac{2w_f}{N_f} \sum_{j=1}^{N_f} r(t_j)\, \hat{u}_\theta(t_j), \qquad \frac{\partial \mathcal{L}}{\partial C} = \frac{2w_f}{N_f} \sum_{j=1}^{N_f} r(t_j)\, \dot{\hat{u}}_\theta(t_j), \qquad \frac{\partial \mathcal{L}_{\text{data}}}{\partial K} = \frac{\partial \mathcal{L}_{\text{data}}}{\partial C} = 0. \tag{3.40}$$

The entire gradient signal driving K and C originates in the physics residual. No part of it derives from fitting the displacement record. This has a direct and testable implication: a network that merely memorises the displacement data, achieving a small $\mathcal{L}_{\text{data}}$ without satisfying the physics, receives no gradient pushing K and C toward their correct values and will leave them at their initialisations. Accurate displacement reproduction is therefore necessary but not sufficient evidence of successful identification, and the parameters must be corroborated independently — which is the purpose of the least-squares and modal comparisons in Sections 5.4 and 5.5.

The gradient expressions also make the observability analysis of Section 3.5 quantitative. The gradient with respect to C is weighted by w_f and, more importantly, its magnitude is proportional to the sensitivity $\partial r/\partial C = \dot{\hat{u}}_\theta$ of the residual to C. Where the damping contribution is small relative to the inertial and elastic terms — which, by the energy argument of Section 3.6, holds whenever $\zeta \ll 1$ over a record short relative to τ — this gradient is correspondingly small, and the convergence of C is slow and loosely constrained. The convergence histories reported in Section 5.6 exhibit exactly this behaviour.

Figure 3.4 renders the complete computational graph, forward and backward. The upper band traces the forward pass: the surrogate is evaluated at collocation points, differentiated twice by automatic differentiation to yield velocity and acceleration, and combined with the known mass and forcing to form the residual of the equation of motion. The lower band traces the backward pass, and it is here that the asymmetry central to this work becomes visible. Gradients originating in the data misfit reach the network weights alone. Gradients originating in the physics residual reach both θ and the identified parameters. There is no path by which fitting the displacement record, on its own, can move K or C.

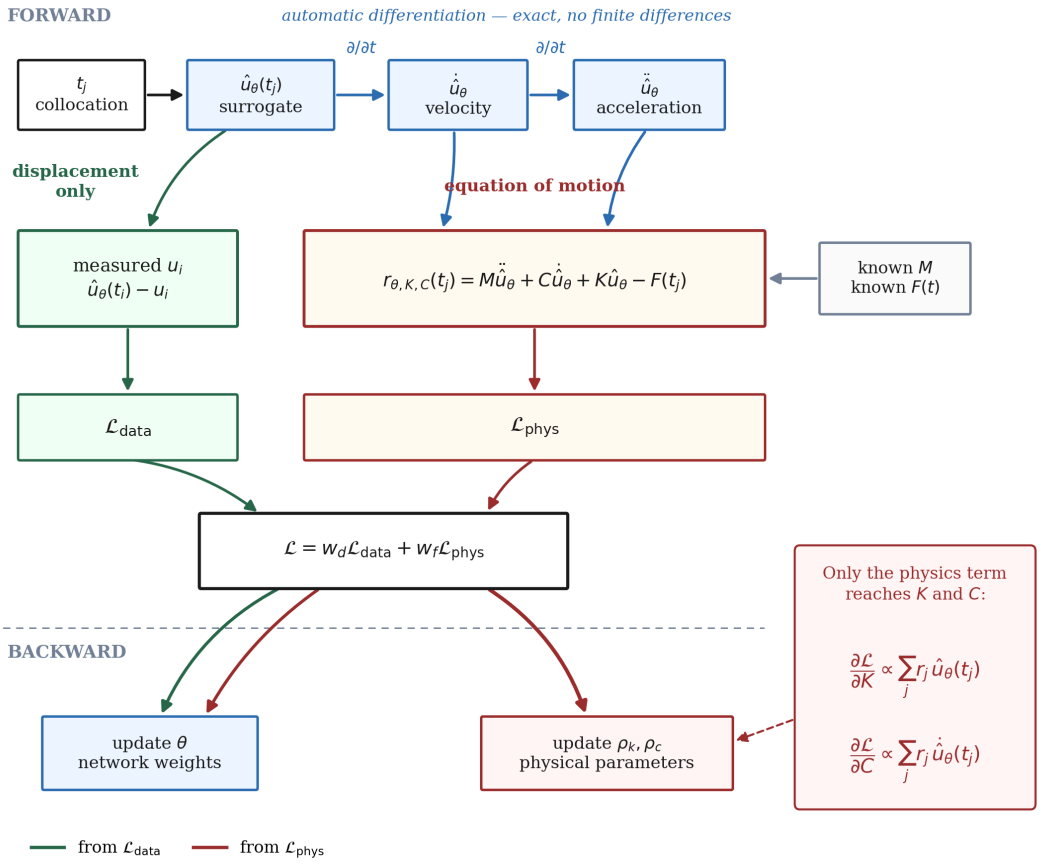


Figure 3.4. Computational graph of the physics-informed identification. Forward (upper band): the surrogate $\hat{u}_\theta(t_j)$ is differentiated with respect to its input by automatic differentiation, exactly and to machine precision, yielding $\dot{\hat{u}}_\theta$ and $\ddot{\hat{u}}_\theta$ without recourse to finite differencing of the measured record. These combine with the fixed generalised mass M and the prescribed forcing $F(t)$ to form the residual r. Backward (lower band): gradients of the composite objective propagate to the trainable quantities. The data misfit reaches only the network weights θ, whereas the physics residual reaches both the weights and the latent parameters (ρ_k, ρ_c). Accurate reproduction of the displacement record is therefore necessary but not sufficient evidence of successful identification, a point examined empirically in Section 5.8.

3.11 Non-Dimensionalisation and Parameter Scaling

The physical quantities of this problem span an extreme range of magnitudes. Displacements are of order 10^{-7} to 10^{-5} m, forces of order 10^4 to 10^5 N, stiffnesses of order 10^8 to 10^{12} N/m, and masses of order 10^4 to 10^7 kg. Optimising directly in these units is numerically hopeless: gradient magnitudes differ across parameters by many orders of magnitude, and a single learning rate cannot serve them all.

The pipeline therefore non-dimensionalises. Introduce reference scales u_{ref} for displacement and F_{ref} for force, and define

$$\hat{u} = \frac{u}{u_{\text{ref}}}, \qquad \hat{F} = \frac{F}{F_{\text{ref}}}. \tag{3.41}$$

Substituting into the equation of motion and dividing by F_{ref},

$$\hat{m}\,\ddot{\hat{u}} + \hat{c}\,\dot{\hat{u}} + \hat{k}\,\hat{u} = \hat{F}(t), \qquad \hat{m} = \frac{M u_{\text{ref}}}{F_{\text{ref}}}, \quad \hat{c} = \frac{C u_{\text{ref}}}{F_{\text{ref}}}, \quad \hat{k} = \frac{K u_{\text{ref}}}{F_{\text{ref}}}, \tag{3.42}$$

so that the trainable quantities are the dimensionless $\hat{k}$ and $\hat{c}$, recovered afterward as

$$K = \hat{k}\,\frac{F_{\text{ref}}}{u_{\text{ref}}}, \qquad C = \hat{c}\,\frac{F_{\text{ref}}}{u_{\text{ref}}}. \tag{3.43}$$

With $u_{\text{ref}} = 10^{-4}$ m and $F_{\text{ref}} = 11{,}103$ N as configured for the X and Y directions (the Z scales are listed in Table 4.3), the dimensionless stiffness falls in the range of order unity to ten, which is precisely the regime in which a single learning rate of 10^{-3} behaves well for both the network weights and the physical parameters. The residual itself is likewise $\mathcal{O}(1)$, so that the data and physics terms of the loss are commensurable and the weighting expresses a genuine modelling preference rather than compensating for unit mismatch.

3.12 Positivity Enforcement by Softplus Reparameterisation

A passive structure dissipates energy, so $C > 0$ is a physical requirement. The unconstrained optimiser respects no such requirement, and — as established in Section 3.5 — an unconstrained estimator in the weakly identifiable regime may drift to negative values. A negative C corresponds to negative dissipation, an energy source rather than a sink, and is inadmissible as a physical damping coefficient.

Positivity is therefore enforced structurally rather than by projection or penalty. Introduce unconstrained latent variables $\rho_k, \rho_c \in \mathbb{R}$ and define

$$\hat{k} = \varsigma(\rho_k), \qquad \hat{c} = \varsigma(\rho_c), \qquad \varsigma(x) = \ln\!\left(1 + e^{x}\right), \tag{3.44}$$

the softplus function. Since ς is smooth, strictly increasing, and strictly positive, the physical parameters are positive by construction for any latent value, and the optimiser may move freely over $\mathbb{R}$. The derivative

$$\varsigma'(x) = \frac{1}{1 + e^{-x}} \in (0, 1) \tag{3.45}$$

is bounded and nonzero for every finite argument, so gradients propagate without saturation over the working range, and the chain rule gives

$$\frac{\partial \mathcal{L}}{\partial \rho_c} = \frac{\partial \mathcal{L}}{\partial \hat{c}}\, \varsigma'(\rho_c). \tag{3.46}$$

Two properties of this construction warrant emphasis. First, it guarantees admissibility of the reported value but does not create information that the data do not contain: in the weakly identifiable regime, the latent ρ_c simply drifts toward $-\infty$, driving $\hat{c}$ toward zero rather than through it. A damping estimate approaching zero with a continuing downward trend in its convergence history is thus the constrained counterpart of the negative estimate returned by an unconstrained method, and carries the same diagnostic meaning. Second, and consequently, the convergence history of $\hat{c}$ must be examined, not merely its final value. A converged parameter exhibits a stable final window; a drifting one does not. Both are reported in Section 5.6.

3.13 Stochastic Gradient Descent and the Adam Update Rule

Optimisation uses Adam (Kingma and Ba, 2015). Its derivation is given here because the convergence diagnostics of Chapter 5 are interpreted in terms of its dynamics.

Let $\mathbf{g}_t = \nabla_\vartheta \mathcal{L}(\vartheta_{t-1})$ denote the gradient at iteration t, where $\vartheta = \{\theta, \rho_k, \rho_c\}$ collects all trainable quantities. Plain gradient descent updates $\vartheta_t = \vartheta_{t-1} - \eta\, \mathbf{g}_t$, which is poorly suited to objectives whose curvature differs sharply across coordinates — exactly the situation here, where network weights and physical parameters have very different sensitivities.

Adam maintains exponential moving averages of the first and second gradient moments:

$$\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1-\beta_1)\, \mathbf{g}_t, \qquad \mathbf{v}_t = \beta_2 \mathbf{v}_{t-1} + (1-\beta_2)\, \mathbf{g}_t^2, \tag{3.47}$$

with $\mathbf{m}_0 = \mathbf{v}_0 = \mathbf{0}$, and by default $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The zero initialisation biases both estimates toward zero in early iterations. To quantify the bias, expand the recursion:

$$\mathbf{m}_t = (1-\beta_1) \sum_{i=1}^{t} \beta_1^{t-i}\, \mathbf{g}_i. \tag{3.48}$$

Taking expectations under the approximation that $\mathbb{E}[\mathbf{g}_i] \approx \mathbb{E}[\mathbf{g}_t]$ over the averaging window,

$$\mathbb{E}[\mathbf{m}_t] \approx \mathbb{E}[\mathbf{g}_t]\, (1-\beta_1) \sum_{i=1}^{t} \beta_1^{t-i} = \mathbb{E}[\mathbf{g}_t]\, (1-\beta_1^t), \tag{3.49}$$

using the geometric sum. The bias-corrected estimates are therefore

$$\hat{\mathbf{m}}_t = \frac{\mathbf{m}_t}{1-\beta_1^t}, \qquad \hat{\mathbf{v}}_t = \frac{\mathbf{v}_t}{1-\beta_2^t}, \tag{3.50}$$

and the update is

$$\vartheta_t = \vartheta_{t-1} - \eta\, \frac{\hat{\mathbf{m}}_t}{\sqrt{\hat{\mathbf{v}}_t} + \epsilon}, \tag{3.51}$$

with $\epsilon = 10^{-8}$ for numerical stability.

The per-coordinate normalisation by $\sqrt{\hat{\mathbf{v}}_t}$ is what makes Adam appropriate here: the effective step size in each coordinate is scaled by the inverse root-mean-square of that coordinate's recent gradients, so that coordinates with persistently small gradients still receive meaningful updates. This is essential for the physical parameters, whose gradients are smaller than those of the network weights by several orders of magnitude even after non-dimensionalisation.

It also has a consequence that must be borne in mind when reading convergence histories. Because the step is normalised by gradient magnitude, a parameter with a very small but consistently signed gradient will continue to move at an appreciable rate rather than stalling. A parameter in the weakly identifiable regime therefore does not freeze — it drifts steadily. Continued drift in a convergence history is thus evidence of weak identifiability, not of an incompletely trained model, and the two must not be conflated. The final-window stability statistics reported in Section 5.6 are computed precisely to distinguish them.

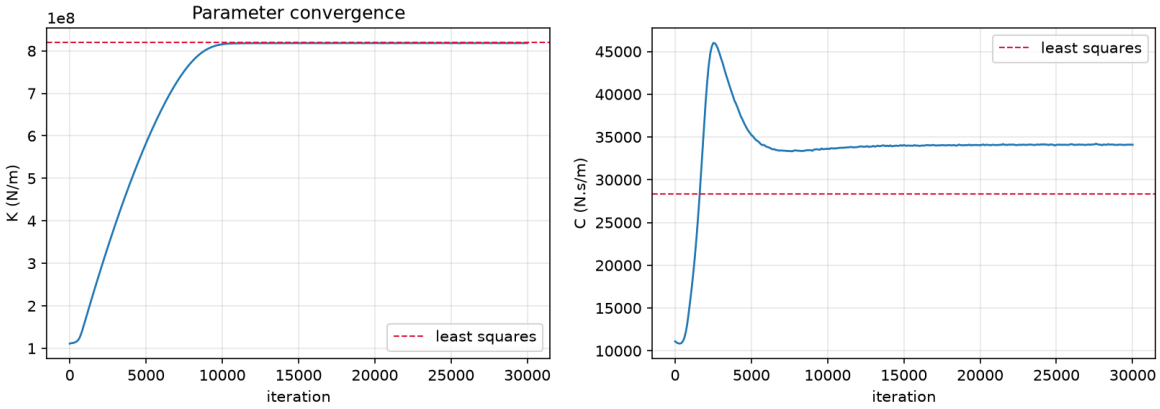


Figure 3.5. Convergence of the identified parameters in the X direction over 30,000 Adam iterations, shown against the independent least-squares reference values.

Chapter 4

Methodology

This chapter documents the computational realisation of the framework derived in Chapter 3. It is written to the standard of reproducibility: a reader with access to the source data should be able to reconstruct every reported result from the description given here. The chapter proceeds from the computing environment, through the software architecture and the direction-agnostic configuration mechanism, to the construction of the reduced-order reference model, the data pipeline, the holdout partition, and the optimiser settings.

4.1 High-Performance Computing Environment

All training runs were executed on a GPU-equipped cluster under a batch scheduler, with each direction submitted as an independent job. Table 4.1 records the execution environment and timing for the three production runs.

Table 4.1. Execution environment and run timings.

Quantity	X	Y	Z
Job identifier	5690	5708	5855
Compute node	gammaweb06	gammaweb06	gammaweb07
GPU hardware	NVIDIA A100-SXM4-40GB (all three runs)		
Devices listed in job inventory	4	4	1
Submission date	9 Sep 2026	9 Sep 2026	13 Sep 2026
Backend	PyTorch	PyTorch	PyTorch
Graph compilation time (s)	74.583	58.376	51.223
Training wall time (s)	251.516	206.759	260.149
Iterations	30,000	30,000	30,000
Exit status	0	0	0

The software stack comprises Python 3.12, PyTorch as the DeepXDE backend, and DeepXDE itself providing the physics-informed problem abstraction. The environment is a project-local virtual environment, pinned and activated by the job script, so that the interpreter and all dependency versions are fixed across runs. Each job writes a header recording the interpreter path, the resolved GPU inventory, and the exact command line, which is retained in the run logs.

Two points of interpretive caution attach to Table 4.1 and are recorded here rather than left implicit. First, the GPU inventory printed in the job header reports the devices *visible* to the job, not the devices actually engaged by the training loop; the X and Y jobs list four devices and the Z job one, but this inventory alone does not establish multi-device execution, and the wall times in the table should not be read as a hardware scaling comparison. Second, the reported training wall time excludes graph compilation, which is listed separately.

Runtime is in any case not the figure of merit for this work. All three runs complete in under five minutes of training on a single A100-class device, which places the method firmly in the regime where identification cost is negligible relative to the finite element analyses that generate the source data.

4.2 Software Architecture

The implementation is organised as an installable Python package `sysid` under `src/`, with thin driver scripts under `scripts/` and a test suite under `tests/`. The separation is deliberate: all physics, data handling, and training logic resides in importable modules that the test suite exercises directly, while the scripts contain only argument parsing and orchestration (Table 4.2).

Table 4.2. Module responsibilities.

Module	Responsibility
`sysid/config.py`	YAML and JSON loading; resolution of the active direction block
`sysid/dataio.py`	CSV ingestion, time windowing, non-dimensionalisation; returns a `Dataset` record
`sysid/physics.py`	Equation of motion, scaling transformations, softplus reparameterisation, residual construction, classical least-squares reference
`sysid/train.py`	Network construction, Fourier encoding, holdout partition, training loop, result serialisation
`sysid/evaluate.py`	Post-training comparison against least-squares and modal references
`sysid/viz.py`	Figure generation

The driver scripts `run_train.py`, `run_evaluate.py`, and `run_plot.py` are executed in sequence, each accepting a `--tag` argument that names the run and determines output filenames. The test suite comprises `test_dataio.py` and `test_physics.py`, which verify the data-loading contract and the scaling and residual transformations respectively; both are executed before any production run.

The data structure returned by ingestion is a frozen record carrying time, physical displacement, scaled displacement, force, the sampling interval, and the parsed metadata and modal reference dictionaries. Downstream code consumes this record rather than reaching back into the CSV, so that the ingestion contract is enforced at a single point.

4.3 The Direction-Agnostic Configuration Mechanism

The central methodological claim of this chapter is that a single unmodified pipeline identifies parameters in all three coordinate directions. The mechanism that realises this is a configuration file containing three mutually exclusive direction blocks, exactly one of which is active at any time. Each block specifies the following groups of settings:

```
data:
  csv: # source record for this direction
  meta: # metadata sidecar
```

```
  modal: # modal analysis reference
  observation_column: # e.g. "Displacement_Z (m)"
  t_min:
  t_max:
physics:
  mass: # generalised mass for the reference mode
  force_amplitude:
  force_frequency:
scaling:
  displacement: # u_ref
  force: # F_ref
network:
  layers:
  activation:
  initializer:
  fourier_features: {enabled:, n:, f_min:, f_max:}
training:
  iterations:
  learning_rate:
  num_domain:
  holdout_every:
  enforce_positive:
  loss_weights:
  k_hat_init:
  c_hat_init:
output:
  dir: # per-direction output subtree
```

Switching direction requires commenting one block and uncommenting another. No source file is edited, and no code path differs between directions. The observation column is read strictly from the active block rather than inferred by pattern matching, so that a misconfigured block fails loudly at load time rather than silently training against the wrong column — an important safeguard when the same CSV carries displacement in several directions.

Output routing is likewise per-direction: results are written to `outputs/phase1/{X,Y,Z}/`, with figures in a `figures/` subdirectory beneath. This prevents overwriting between concurrently scheduled cluster jobs and makes the provenance of every artefact unambiguous from its path alone. Table 4.3 records the configuration actually used for each production run.

Note on the encoding band. The Z direction uses a wider Fourier band, 5–60 Hz rather than 5–40 Hz, because its reference mode lies at 51.43 Hz — above the upper edge of the band adequate for X and Y, whose reference modes lie near 20 Hz. This is a direct application of the design rule derived in Section 3.9, that the encoding band must bracket the spectral content of the target response, and it is the one encoding quantity that necessarily differs between directions on physical grounds.

The input layer width of 256 is not specified by hand: the configuration requests a single time input, and the training routine overwrites `layers[0]` with $2m$ when Fourier encoding is enabled, since the encoding emits a sine and a cosine channel per frequency. The hidden

Table 4.3. Active configuration by direction.

Setting	X	Y	Z
Observation column	`Displacement_X (m)`	`Displacement_Y (m)`	`Displacement_Z (m)`
Reference mode	1	2	4
Generalised mass (kg)	56,767.0	57,485.7	5,736,762.0
Force amplitude F_0 (N)	11,103.0	11,103.0	95,073.0
Force frequency (Hz)	15.0	15.0	15.0
Displacement scale u_{ref} (m)	10^{-4}	10^{-4}	2×10^{-7}
Force scale F_{ref} (N)	11,103.0	11,103.0	95,073.0
Samples	1,000	1,000	10,000
Time span (s)	0.01–10.00	0.01–10.00	0.001–10.00
Sampling interval (s)	0.01	0.01	0.001
Sampling / Nyquist (Hz)	100 / 50	100 / 50	1000 / 500
Collocation points	10,000	10,000	10,000
Fourier frequencies m	128	128	128
Fourier band (Hz)	5–40	5–40	5–60
Network layers	[256, 64, 64, 64, 64, 1]	[256, 64, 64, 64, 64, 1]	[256, 64, 64, 64, 64, 1]

architecture is four layers of 64 units with sinusoidal activation and Glorot uniform initialisation, identical across directions.

4.4 The Reduced-Order Reference Model

Section 3.3 derived the transformation from modal to physical-observation coordinates,

$$M_u = \frac{M_r}{\phi_{r,j}^2}, \qquad K_u = \frac{K_r}{\phi_{r,j}^2}, \tag{4.1}$$

and established that these generalised quantities are coordinate-dependent. The metadata sidecar accompanying each direction records the selected reference mode, its natural frequency, the mode-shape component at the observation node, and the resulting generalised mass and stiffness. These values may be checked against the derivation directly, and Table 4.4 reports that check.

The identities $M_u = 1/\phi_{r,j}^2$ and $K_u = (2\pi f_r)^2 M_u$ are satisfied to full recorded precision in both the Y and Z directions. The first identity confirms that the modal vectors supplied by the finite element modal analysis are mass-normalised, so that $M_r = 1$ and M_u reduces to the reciprocal squared mode-shape component at the observation node.

This verification carries real interpretive weight, and it resolves what would otherwise be the most conspicuous anomaly in the dataset. The Z generalised mass of 5,736,762 kg exceeds the total structural mass of the table, 78,500 kg, by a factor of approximately 73. Taken at face value this is absurd. Understood through the transformation it is exactly correct: the mode-shape component at Node 3575 is $\phi_{4,j} = 4.175 \times 10^{-4}$ in Z, an order of magnitude

Table 4.4. Verification of the modal reduction identities.

Quantity	X	Y	Z
Reference mode r	1	2	4
f_r (Hz)	20.0835	20.65810213	51.43420362
$\phi_{r,j}$ at Node 3575	—	4.17081×10^{-3}	4.17510×10^{-4}
$1/\phi_{r,j}^2$ (kg)	—	57,485.71703	5,736,762.05033
Recorded M_u (kg)	56,767.38892	57,485.71703	5,736,762.05033
$(2\pi f_r)^2 M_u$ (N/m)	9.03936×10^8	9.68502×10^8	5.99143×10^{11}
Recorded K_u (N/m)	9.03936×10^8	9.68502×10^8	5.99143×10^{11}

The X sidecar does not record $\phi_{r,j}$, so the first identity cannot be evaluated for X (—).

smaller than the component $\phi_{2,j} = 4.171 \times 10^{-3}$ of the Y reference mode, and since M_u scales as $\phi_{r,j}^{-2}$, a mode-shape component ten times smaller produces a generalised mass one hundred times larger. Node 3575 simply lies close to a nodal region of mode 4 in the Z direction. The generalised mass is a coordinate-dependent scaling factor, not a physical mass, and the magnitudes reported across the three directions are precisely what the theory of Section 3.3 predicts.

The same reasoning explains the stiffness magnitudes. The Z generalised stiffness of 5.99×10^{11} N/m is not a statement that the table is three orders of magnitude stiffer vertically; it is ω_r^2 times a generalised mass inflated by the same factor. The frequency ratio $K_u/M_u = \omega_r^2$ is the invariant, and the identified natural frequency is therefore the quantity that may be compared meaningfully across directions.

Note that the configuration files use rounded generalised masses — 56,767.0 against the exact 56,767.38892, and 57,485.7 against 57,485.71703 — introducing relative discrepancies of 6.9×10^{-6} and 3.0×10^{-7} respectively. These are far too small to affect any reported comparison and are recorded only for completeness.

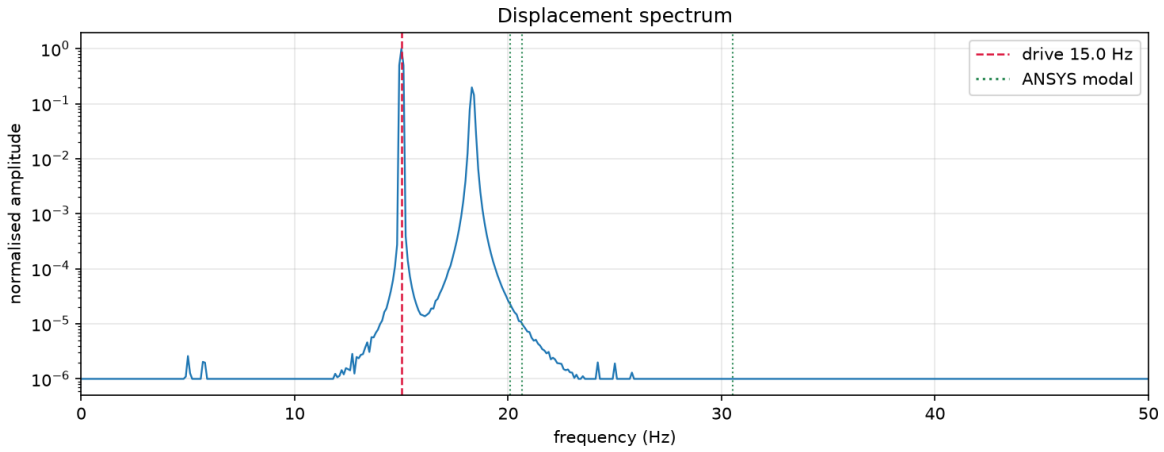


Figure 4.1. Response spectrum for the Y direction, showing the drive component, the structural response content near 18.3 Hz, and the modal reference line for mode 2.

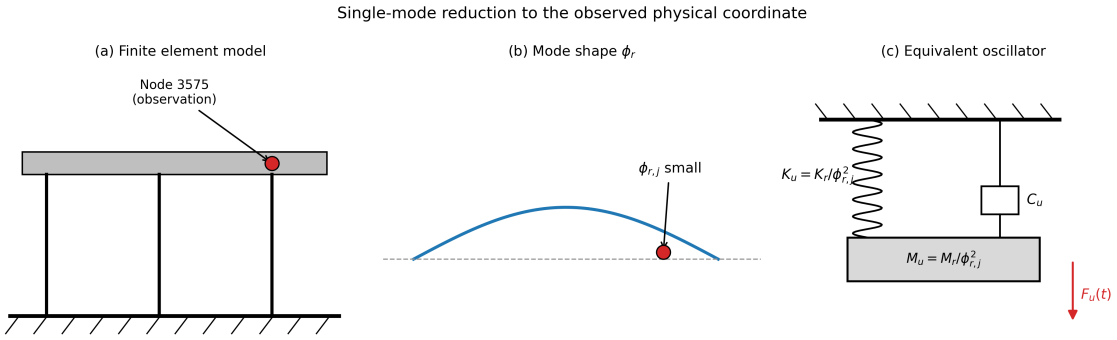


Figure 4.2. Single-mode reduction. A small mode-shape component at the observation node produces a large generalised mass through the $\phi_{r,j}^{-2}$ scaling, as observed in the Z direction.

4.5 Data Ingestion and the Displacement-Only Decision

Each source CSV carries time, displacement, velocity, acceleration, and force columns exported from the transient finite element analysis. The ingestion routine locates the time and force columns by tolerant case-insensitive keyword matching, so that a re-exported record with slightly altered headers still loads, while the displacement column is taken strictly from the active configuration block for the reason given in Section 4.3.

The routine then applies the configured time window, computes the median sampling interval, and checks uniformity, issuing a warning if the spread of successive intervals exceeds a fixed relative tolerance of the median. All three records pass this check with strictly increasing timestamps and no non-finite entries. A record reduced below fifty samples by an over-narrow window raises an error rather than proceeding.

Only the displacement column is used as a training observation. This is the single most consequential methodological decision in the pipeline, and it is made deliberately rather than by default. The velocity and acceleration columns exported alongside the displacement are not consistent numerical derivatives of it: the correlation between the exported acceleration and the numerically differentiated velocity is 0.986 in Z and 0.88 in X and Y. Fitting against those columns produces negative — that is, unphysical — damping estimates. The discrepancy reflects differences in the differentiation scheme, the integration scheme, and the output sampling convention of the finite element solver, and it is not in itself evidence that any column is wrong. But it does mean that the exported kinematic columns cannot be treated as a mutually consistent set.

The PINN sidesteps the problem entirely. It observes displacement alone and obtains velocity and acceleration by automatic differentiation of its own surrogate, which is internally consistent by construction. This is precisely the structural advantage over classical estimators identified in Section 1.2, and here it is not a theoretical advantage but a decisive practical one: the classical estimator has no alternative but to differentiate something, whereas the PINN differentiates a function it controls. The exported velocity and acceleration columns are retained and used only as diagnostic cross-checks, never as training targets.

4.6 Non-Dimensionalisation as Implemented

The scaling of Section 3.11 is applied at ingestion for the displacement and inside the residual for the force. Ingestion divides the measured displacement by u_{ref} to produce the dimensionless $\hat{u} = u/u_{\text{ref}}$ that the network targets. The residual constructs its forcing term as

$$\hat{F}(t) = \frac{F_0}{F_{\text{ref}}} \sin(2\pi f_0 t), \tag{4.2}$$

and the scaled mass is computed once as $\hat{m} = M u_{\text{ref}}/F_{\text{ref}}$. Recovery to physical units is a single multiplication, $K = \hat{k}\, F_{\text{ref}}/u_{\text{ref}}$, and likewise for C. The forward and inverse transformations are implemented as a matched pair in `physics.py` and are exercised directly by the test suite, which verifies round-trip consistency.

Two implementation details deserve explicit statement because they bear on the interpretation of the residual.

Time is not rescaled. Derivatives in the residual are taken with respect to real time in seconds. Only displacement and force are non-dimensionalised. This keeps the identified parameters in recoverable physical units without a time-scaling factor propagating through the derivative terms.

The residual forcing is the configured analytic harmonic. The physics residual evaluates $\hat{F}(t)$ from the configured amplitude and frequency as a pure sine, rather than interpolating the force column of the CSV. The collocation points at which the residual is imposed are in general not sample times, so an analytic force is the natural choice; but it does mean that the right-hand side seen by the PINN is the configured harmonic, and that any difference between that harmonic and the recorded force is not represented in the physics term. The configured amplitudes are those given in Table 4.3. This distinction matters when comparing against the classical estimator, which by contrast uses the recorded force column directly, and it is handled explicitly in Section 4.10. Table 4.5 records the resulting dimensionless magnitudes.

Table 4.5. Dimensionless parameter magnitudes at convergence.

Quantity	X	Y	Z
$\hat{k}$ identified	7.37046	7.92845	1.26056
$\hat{c}$ identified	3.071×10^{-4}	3.097×10^{-4}	4.951×10^{-9}
Recovery factor $F_{\text{ref}}/u_{\text{ref}}$ (N/m)	1.1103×10^{8}	1.1103×10^{8}	4.7537×10^{11}

The identified $\hat{k}$ values lie between 1 and 8 across all three directions — the regime for which the learning rate was chosen, confirming that the scaling achieves its purpose.

4.7 Collocation and Holdout Construction

Collocation. The time domain is declared as the interval $[t_{\min}, t_{\max}]$ spanned by the loaded record, and 10,000 interior collocation points are sampled within it, with two boundary points. The retained training sample times are additionally supplied as anchors, guaranteeing that the residual is evaluated at every observed time as well as at the sampled interior points.

Collocation density relative to the oscillation period is the meaningful measure. At a 15 Hz drive over a 10 s window, the record spans 150 drive cycles, giving roughly 67 collocation points per drive cycle — comfortably above the density at which the residual can resolve the oscillation. An earlier configuration used 4,000 collocation points, yielding approximately 27 per cycle, and was increased to the present value during development.

Holdout. DeepXDE's reported test loss is evaluated on points drawn from the training distribution and therefore does not measure generalisation. An explicit holdout partition is constructed instead, by the deterministic rule

$$\text{sample } i \text{ is held out} \iff i \bmod 5 = 2, \tag{4.3}$$

with the remainder forming the training set. This withholds exactly 20 % of samples: 200 of 1,000 in X and Y, and 2,000 of 10,000 in Z.

Two properties of this rule matter. First, it is deterministic, so the partition is identical across runs and seeds and the comparison between them is exact. Second, the held-out points are distributed uniformly through the record rather than taken as a terminal block, so the validation covers the entire time span rather than a single interval. Held-out samples are excluded from the data loss entirely and are never presented to the network during training.

One limitation of this construction is recorded plainly, since it governs what the held-out metrics may be taken to establish. Because the held-out points are interleaved with training points at a spacing of five samples, the holdout measures *interpolation within the record*. It is not a forecast, a different node, a different loading case, or an independent experiment. It establishes that the surrogate is not memorising individual samples — a real and necessary check, given the gradient structure of Section 3.10 — but it does not by itself validate the identified physical parameters. That validation comes from the independent comparisons of Section 4.10, and the distinction is maintained throughout the results.

4.8 Network Construction and Fourier Encoding

The network is a fully connected feedforward architecture constructed through DeepXDE's `FNN` class with sinusoidal activation and Glorot uniform initialisation, with a feature transform applied to the input.

When Fourier encoding is enabled, the transform maps a scalar time to a $2m$-dimensional

vector,

$$t \mapsto \gamma(t) \in \mathbb{R}^{2m}, \qquad b_i = f_{\min} + (i-1)\,\frac{f_{\max} - f_{\min}}{m-1}, \quad i = 1, \ldots, m, \tag{4.4}$$

with the frequencies spaced linearly across the configured band. The input layer width is set programmatically to $2m = 256$. Note that this is a deterministic linear sweep rather than a random draw from a distribution; the construction is that of Tancik et al. (2020) with a deterministic rather than stochastic frequency set, which removes one source of run-to-run variation.

The implementation carries an explicit diagnostic for the spectral-bias failure mode derived in Section 3.8. When Fourier encoding is disabled, the training routine computes the number of oscillations spanned by the window, $N_{\text{osc}} = f\,(t_{\max} - t_{\min})$, and issues a warning when $N_{\text{osc}} > 30$, directing the user either to shorten the window or to enable the encoding. This threshold was established empirically during development and matches the theoretical prediction: with Glorot-scale initialisation the first layer of a raw-coordinate network spans approximately one cycle across the input range, and gradient descent does not climb from there to 190 cycles. The characteristic failure is not graceful degradation but collapse to the zero-output solution, which is a stable local minimum and presents as a flat, stalled loss curve.

All three production runs use Fourier encoding. The encoding supplies the frequencies; the network learns only the amplitudes.

4.9 Optimiser Configuration and Positivity Enforcement

Training uses Adam at a fixed learning rate of 10^{-3} for 30,000 iterations, with loss weights $w_f = 1$ and $w_d = 100$ applied to the physics-residual and data terms respectively, so that the network is anchored firmly to the measurement. No learning-rate schedule and no second-stage L-BFGS refinement are applied; the reported results are those of the Adam run alone.

The two physical parameters are registered as external trainable variables and optimised jointly with the network weights in the same backward pass, exactly as derived in Section 3.10.

Positivity. The softplus reparameterisation of Section 3.12 is implemented by storing the raw latent values ρ_k, ρ_c as the trainable variables and applying ς inside the residual. The initialisation deserves note: the configuration specifies the desired starting values in scaled units, $\hat{k}_{\text{init}} = 1.0$ and $\hat{c}_{\text{init}} = 10^{-4}$ (the latter for X and Y; the Z run starts C near 9.5×10^4 N·s/m), and the routine applies the inverse softplus

$$\rho = \varsigma^{-1}(y) = \ln(e^y - 1), \tag{4.5}$$

computed in the numerically stable `log(expm1(y))` form with a linear passthrough for large y, so that the configured value means what it says after the transform. This preserves the interpretability of the configuration file under the reparameterisation.

The choice is consequential and was made in response to observed behaviour: unconstrained

runs on this dataset were seen to drive C to negative values during training before recovering. The softplus construction makes negative values unreachable rather than merely penalised.

Initialisation distance. The initial $\hat{k} = 1.0$ is far from the converged X and Y values of 7.37 and 7.93. In physical units, K begins at 1.110×10^8 N/m in X and Y and at 4.754×10^{11} N/m in Z. That the parameters travel this distance is itself evidence against memorisation: a network that fitted the displacement without satisfying the physics would receive no gradient on $\hat{k}$ and would leave it at its initialisation, which is exactly the signature of the failed development runs. In Z the converged value of 1.26 lies closer to the start, and the evidence against memorisation there rests chiefly on the independent comparisons of Chapter 5.

Convergence logging. A callback records both raw latent parameters every 100 iterations to a `.dat` file, giving 301 logged states from iteration 0 to 30,000 inclusive. A post-processing routine applies softplus and the physical recovery factor to produce a tidy CSV in engineering units. The softplus is applied in the `log1p(exp(v))` form with a linear passthrough above 30 to avoid overflow, matching the forward transform used during training. These convergence records are the basis for the final-window stability statistics reported in Chapter 5, and it is worth emphasising why they are logged at all: as established in Section 3.13, Adam's per-coordinate gradient normalisation means that a weakly constrained parameter drifts steadily rather than stalling, so the trajectory of a parameter carries information that its final value does not.

4.10 The Classical Least-Squares Reference

The independent comparison estimator implements the formulation of Section 1.2 with spectral differentiation. Given the displacement record, the routine computes the real FFT $\tilde{u}(\omega)$, multiplies by $\mathrm{i}\omega$ and $-\omega^2$ to obtain velocity and acceleration spectra, and inverts:

$$\dot{u}(t) = \mathcal{F}^{-1}\big[\mathrm{i}\omega\, \tilde{u}(\omega)\big], \qquad \ddot{u}(t) = \mathcal{F}^{-1}\big[-\omega^2\, \tilde{u}(\omega)\big]. \tag{4.6}$$

This is exact for a band-limited periodic signal and is preferred over the CSV's exported velocity and acceleration columns for the consistency reasons of Section 4.5.

Spectral differentiation assumes periodicity, so the record endpoints are contaminated by wraparound. The routine trims n_{trim} samples from each end — 50 per side for the 1,000-sample X and Y records, 500 per side for the 10,000-sample Z record — leaving 900, 900, and 9,000 rows respectively for the regression. The design matrix and right-hand side are assembled as in Equation (1.3),

$$\mathbf{A} = \big[\dot{u} \;\; u\big], \qquad \mathbf{b} = F - M\ddot{u}, \tag{4.7}$$

and solved in the least-squares sense, returning C, K, the coefficient of determination R^2 of the regression, and classical standard errors from $\hat{\sigma}^2(\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}$.

Three interpretive cautions attach to this reference and are stated here so that Chapter 5 may rely on them.

1. *The regression R^2 describes the fit to $F - M\ddot{u}$*, not to displacement and not to force. A high value indicates that the two-column model explains the variance of that particular target; it is not interchangeable with a displacement-fit quality, and the residual-to-force ratio is reported alongside it as the more informative diagnostic.

2. *The standard errors assume an adequate linear model with independent, homoscedastic residuals.* Serial correlation, spectral differentiation error, endpoint leakage, and structural-reduction error all violate these assumptions to some degree. The reported standard errors are therefore indicative of estimator precision under an idealisation, not validated confidence bounds on the physical parameters.

3. *The force convention must be matched for the comparison to be meaningful.* As noted in Section 4.6, the PINN residual uses the configured analytic harmonic while the least-squares routine uses the recorded force column. Where these differ, the two estimators are solving different problems and their outputs are not directly comparable. Chapter 5 therefore reports the least-squares reference under both conventions — the recorded force column, and the same analytic harmonic the PINN uses — and the same-force comparison is the one on which any claim of agreement between the two methods rests.

4.11 Reproducibility Controls

The following controls fix the reported results.

Seeding. The DeepXDE global random seed is set explicitly at the start of every run, seeding the Python, NumPy, and PyTorch generators. Production runs use seed 0. A repeat at seed 7 was performed in the X direction during development to assess initialisation sensitivity, and its outcome is reported in Section 6.2.

Determinism of the partition. The holdout rule is arithmetic on the sample index and carries no randomness, so the partition is byte-identical across runs.

Determinism of the encoding. The Fourier frequencies are a linear sweep, not a random draw, so the input encoding is identical across runs.

Environment pinning. A project-local virtual environment fixes interpreter and dependency versions; the job header records the resolved interpreter path.

Artefact provenance. Every run writes its identified parameters, fixed inputs, derived quantities, scaled parameters, run metadata, fit metrics, and validation metrics to a single JSON keyed by run tag, alongside the per-sample prediction CSV and the convergence history. Outputs are routed by direction, so the source of every artefact is determined by its path.

Pre-run testing. The `pytest` suite is executed before production runs, verifying the data-loading contract and the scaling and residual transformations.

Table 4.6 summarises the complete specification.

Table 4.6. Consolidated methodological specification.

Category	Specification
Observation	Displacement only, single node (3575), single direction per run
Surrogate	FNN, 4 hidden layers × 64 units, sine activation, Glorot uniform
Input encoding	128 linearly spaced Fourier frequency pairs, input dimension 256
Derivatives	Automatic differentiation of the surrogate
Trainable	Network weights θ, $\hat{k}$, $\hat{c}$ (softplus-transformed)
Loss	$\mathcal{L} = w_d\mathcal{L}_{\text{data}} + w_f\mathcal{L}_{\text{phys}}$, weights $w_d = 100$, $w_f = 1$
Collocation	10,000 interior points + 2 boundary + training-time anchors
Holdout	Deterministic 20 %, $i \bmod 5 = 2$, uniformly distributed
Optimiser	Adam, $\eta = 10^{-3}$, 30,000 iterations, no schedule
Scaling	$u_{\text{ref}} = 10^{-4}$ m, $F_{\text{ref}} = 11{,}103$ N (X, Y; Z in Table 4.3); time unscaled
Positivity	Softplus reparameterisation with inverse-softplus initialisation
Logging	Parameters every 100 iterations, 301 states per run
Reference 1	Least squares with spectral derivatives, 5 % endpoint trim, both force conventions
Reference 2	ANSYS modal analysis, selected reference mode per direction
Seed	0 (production); 7 (sensitivity check, X)

Chapter 5

Results and Discussion

This chapter presents the outcome of the three production runs. It proceeds from the accuracy of the displacement surrogate, through the identified parameters and their comparison against two independent references, to the convergence behaviour of the optimisation and the window-length study that explains it. The organisation is deliberate: surrogate accuracy is a necessary precondition for identification but, as established by the gradient structure of Section 3.10, not sufficient evidence of it, and the two must be assessed separately.

5.1 Overview

All three directions were trained with the identical pipeline, architecture, loss, and optimiser documented in Chapter 4, switched by configuration alone. All three runs completed without error in under 4.5 minutes of training. Figure 5.1 summarises the pipeline that produced these results.

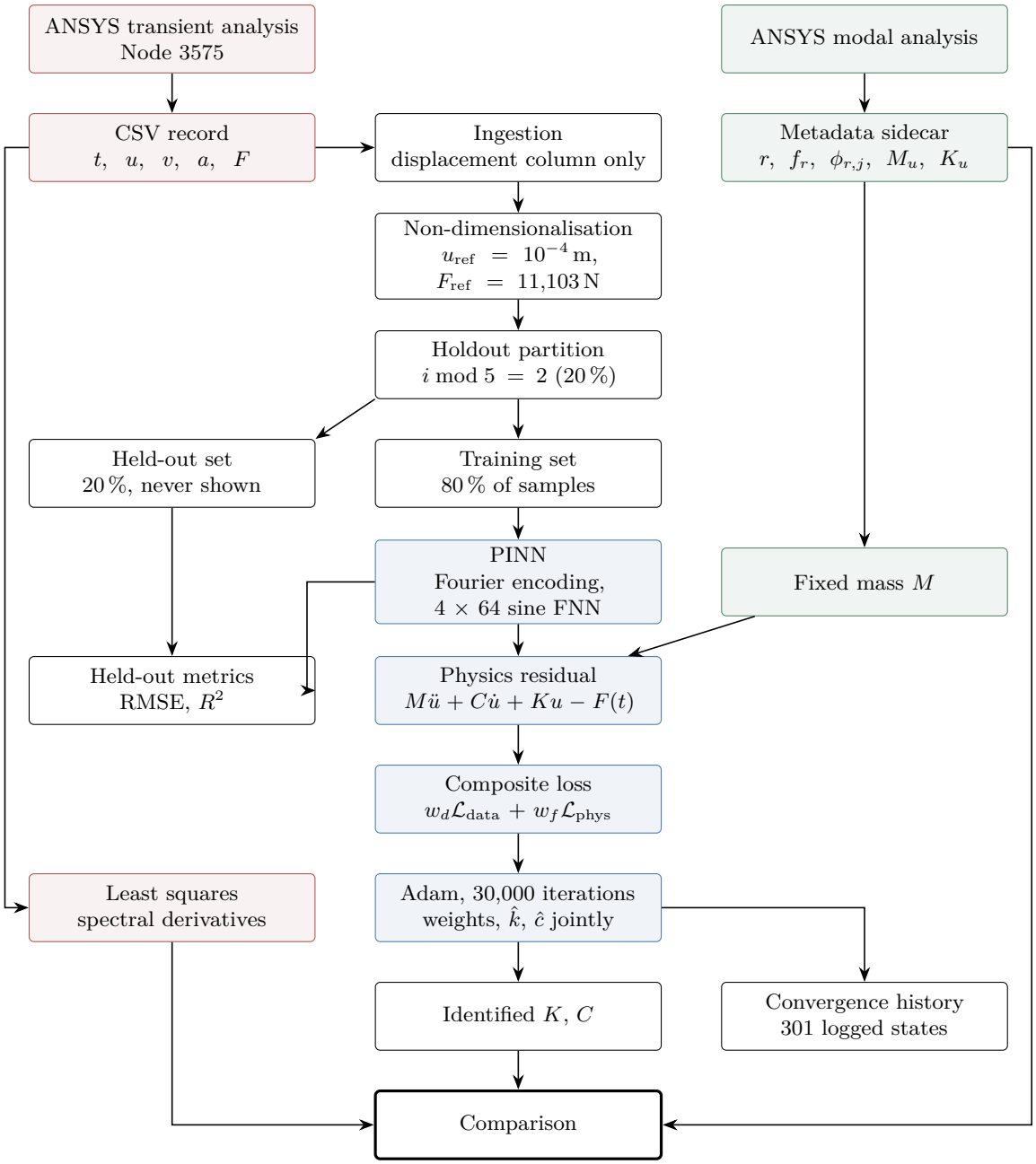


Figure 5.1. The identification pipeline, from finite element output to comparison against independent references. The scale values shown are those of the X and Y runs.

5.2 Surrogate Accuracy and Generalisation

The displacement surrogate reproduces the measured record to high accuracy in every direction. Table 5.1 reports the fit over all samples and, separately, over the held-out partition that was never presented to the network.

Table 5.1. Surrogate accuracy and generalisation.

Metric	X	Y	Z
Samples, total	1,000	1,000	10,000
Samples, training / held out	800 / 200	800 / 200	8,000 / 2,000
RMSE, all samples (m)	2.025×10^{-7}	**[Y RMSE]**	4.043×10^{-10}
Relative RMSE, all samples	0.797 %	0.785 %	0.327 %
Maximum absolute error (m)	**[X]**	**[Y]**	**[Z]**
R^2, all samples	0.99993648	0.99993833	0.99998933
RMSE, training (m)	$\approx 1.97 \times 10^{-7}$	**[Y]**	$\approx 4.05 \times 10^{-10}$
RMSE, held out (m)	$\approx 2.25 \times 10^{-7}$	**[Y]**	$\approx 3.97 \times 10^{-10}$
Held-out / training RMSE	1.14	1.51	0.98
Relative RMSE, held out	0.884 %	1.059 %	0.321 %
R^2, held out	0.99992176	0.99988774	0.99998972

Three observations follow.

Generalisation is essentially perfect in all three directions. The held-out coefficients of determination exceed 0.9998 everywhere, and the ratio of held-out to training RMSE is 1.14 in X, 1.51 in Y, and 0.98 in Z. A network that had memorised individual samples would show held-out error dramatically exceeding training error; none does. In Z the held-out RMSE is marginally below the training RMSE, which is within the variation expected between two subsets of a smooth record.

The Z surrogate is the most accurate of the three, at 0.327 % relative RMSE against 0.797 % and 0.785 %. This is attributable to its sampling density: the Z record carries 10,000 samples at 1 kHz, giving 66.7 samples per drive cycle against 6.7 in X and Y. The Fourier-encoded network has correspondingly more information per cycle to fit.

These metrics establish interpolation quality, not parameter validity. As set out in Section 4.7, the held-out points are interleaved with training points at a spacing of five samples, so the holdout measures interpolation within the record. It confirms that the surrogate is not memorising — a real and necessary check given the gradient structure of Section 3.10 — but it says nothing directly about whether K and C are correct. That question is addressed in Sections 5.4 and 5.5.

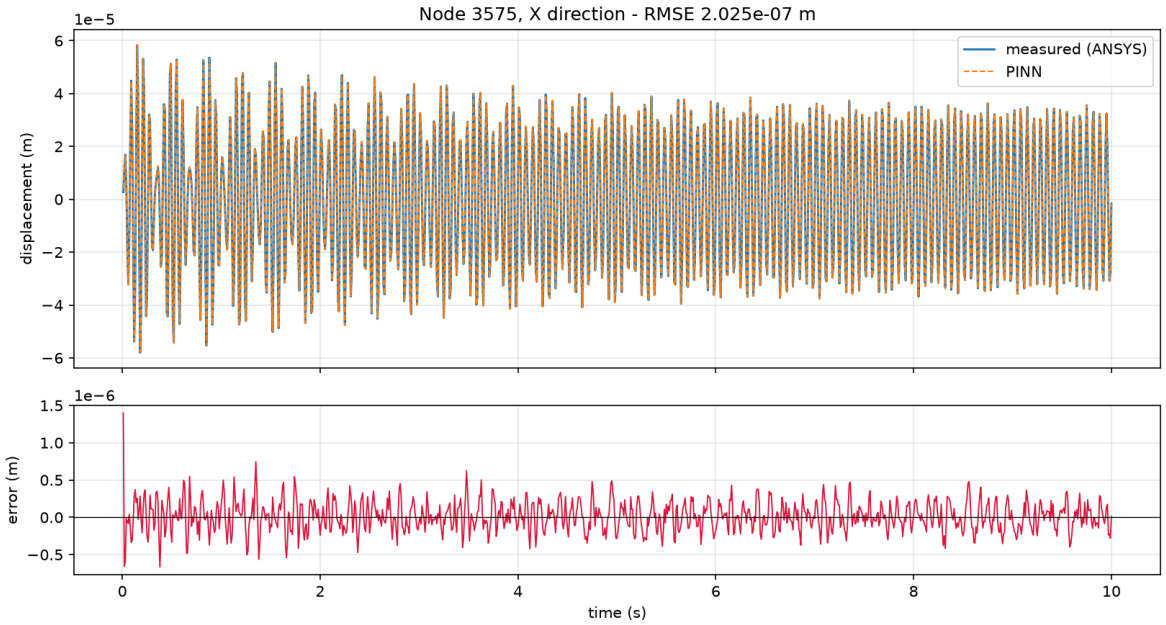


Figure 5.2. Predicted against measured displacement, X direction, over the full 10 s record (150 drive cycles), with error panel.

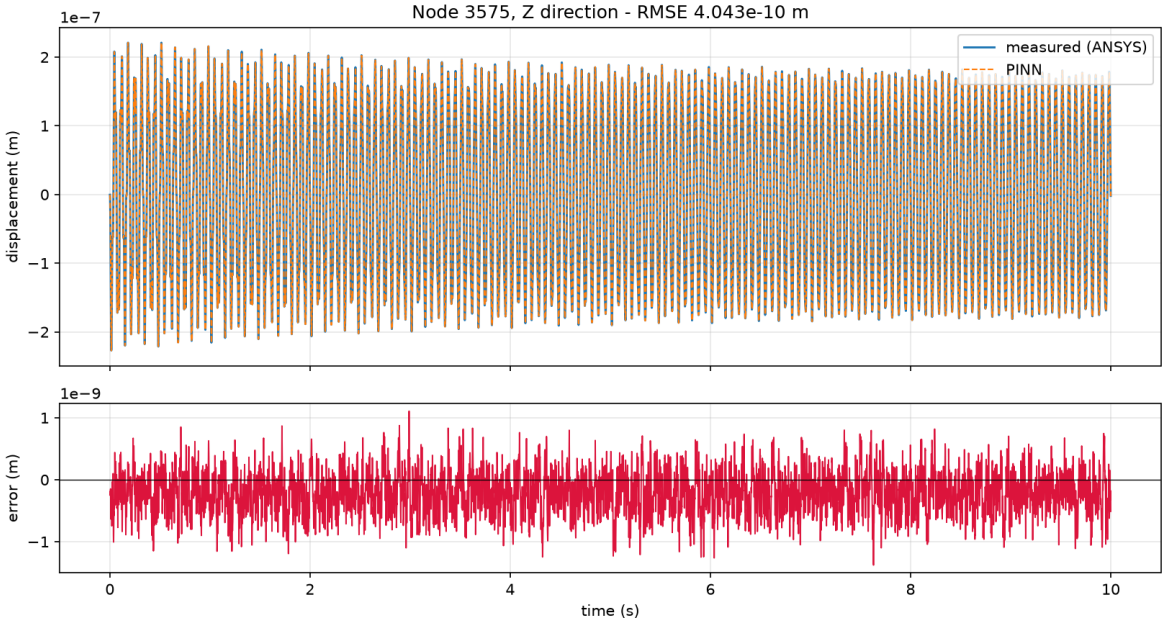


Figure 5.3. Predicted against measured displacement, Z direction. The higher sampling density is visible in the resolution of the trace.

5.3 Identified Parameters

Table 5.2 reports the identified parameters and the quantities derived from them through Equations (3.12) and (3.18).

Table 5.2. Identified parameters and derived quantities.

Quantity	X	Y	Z
Reference mode	1	2	4
Fixed mass M (kg)	56,767.0	57,485.7	5,736,762.0
Identified K (N/m)	8.1834×10^8	8.8030×10^8	5.9923×10^{11}
Identified C (N·s/m)	3.4093×10^4	3.4381×10^4	2.3534×10^3
Scaled $\hat{k}$	7.37046	7.92845	1.26056
Scaled $\hat{c}$	3.071×10^{-4}	3.097×10^{-4}	4.951×10^{-9}
ω_n (rad/s)	120.0659	123.7469	323.1931
f_n (Hz)	19.1091	19.6949	51.4378
ζ (%)	0.2501	0.2417	0.0000635
Decay time constant τ (s)	3.330	3.344	4,875.4

The X and Y damping ratios of 0.250 % and 0.242 % are physically plausible for a welded steel structure, falling within the range typically quoted for such systems. The Z damping ratio of 6.35×10^{-5} % is four orders of magnitude smaller and implies a decay time constant of 4,875 seconds. The significance of that last row is developed in Section 5.7.

5.4 Comparison Against the Classical Least-Squares Reference

The classical estimator provides the primary independent check. As established in Section 4.10, the comparison is meaningful only when both estimators use the same right-hand side, so Table 5.3 reports the least-squares reference under both force conventions: the recorded force column, and the analytic harmonic that the PINN residual actually uses.

Table 5.3. Least-squares reference under both force conventions.

Direction / convention	K (N/m)	C (N·s/m)	Regression R^2	Residual / F RMS
X / recorded force	8.206×10^8	$\approx 2.83 \times 10^4$	0.998561	10.009 %
X / analytic harmonic	8.206×10^8	$\approx 2.83 \times 10^4$	0.998561	10.007 %
Y / recorded force	8.836×10^8	$\approx 3.25 \times 10^4$	0.998379	9.657 %
Y / analytic harmonic	8.836×10^8	$\approx 3.25 \times 10^4$	0.998380	9.656 %
Z / recorded force	1.159×10^9	$\approx -6.1 \times 10^5$	0.000209	135.793 %
Z / analytic harmonic	$\approx 5.991 \times 10^{11}$	**[negative]**	0.998756	3.888 %

In X and Y the two conventions differ only in the sixth significant figure, so the displayed values coincide.

The convention matters little in X and Y, where the recorded force and the configured harmonic differ by only 10 N RMS, changing K in the sixth significant figure. It matters decisively in Z, where the two conventions differ by 74,520 N RMS and produce stiffness estimates separated by

a factor of 517. Table 5.4 gives the resulting agreement, using the matched-force comparison as the meaningful one.

Table 5.4. PINN against the classical least-squares reference.

Comparison	X	Y	Z
K, PINN vs recorded-force LS	−0.28 %	−0.38 %	factor 517
K, PINN vs matched-force LS	−0.28 %	−0.38 %	0.015 %
C, PINN vs matched-force LS	+20.4 %	≈ +6 %	not comparable (LS $C < 0$)
LS standard error of C	38.3 %	38.1 %	—

The stiffness result is the principal finding of this chapter, and it is strong in all three directions. Under a matched force convention, two methods sharing no machinery — one a neural surrogate trained by stochastic gradient descent on displacement alone, the other a closed-form linear regression on spectrally differentiated data — agree on the identified stiffness to 0.28 %, 0.38 %, and 0.015 %. The methods differ in their derivative estimation, their treatment of endpoints, their optimisation, and their numerical conditioning. Agreement at this level across three directions and three orders of magnitude in stiffness magnitude is substantial corroboration that the identified stiffnesses are correct.

The damping comparison is weaker, and differs in character between directions. In X and Y the PINN exceeds the matched-force least-squares damping by 20.4 % and approximately 6 %. These are real discrepancies, but both estimates are positive, of the same order, and correspond to damping ratios in the same physically plausible band. Their magnitude should be read against the least-squares standard errors, which are 38.3 % and 38.1 % of the respective damping estimates: the discrepancies sit within the classical estimator's own precision. In Z the matched-force least-squares damping is negative, and thus inadmissible as a physical damping coefficient for a passive structure. Section 5.7 explains why.

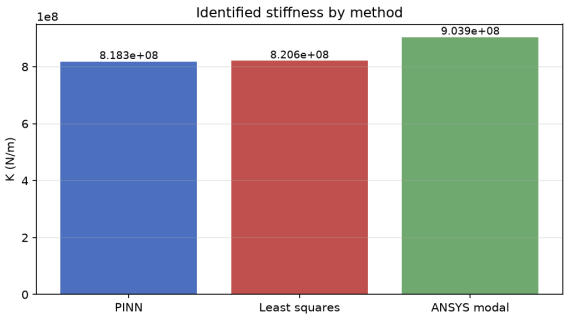


Figure 5.4. Identified stiffness by method, X direction.

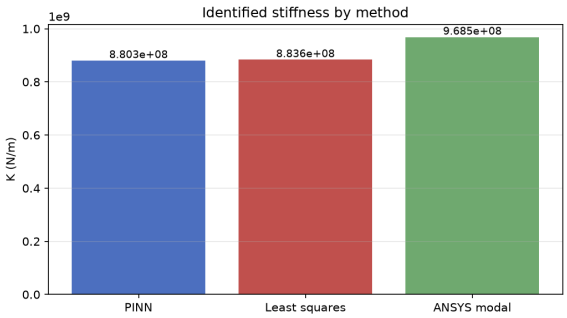


Figure 5.5. Identified stiffness by method, Y direction.

5.5 Comparison Against the Modal Reference

The finite element modal analysis provides a second, wholly independent reference. Table 5.5 reports it.

Table 5.5. PINN against the modal reference.

Quantity	X	Y	Z
Modal K_u (N/m)	9.039×10^8	9.685×10^8	5.991×10^{11}
Modal f_r (Hz)	20.0835	20.6581	51.4342
PINN K (N/m)	8.183×10^8	8.803×10^8	5.992×10^{11}
PINN f_n (Hz)	19.1091	19.6949	51.4378
K difference	−9.47 %	−9.11 %	+0.014 %
f difference	−4.85 %	−4.66 %	+0.007 %

The X and Y directions show a consistent stiffness deficit of approximately 9 % against the modal reference. The consistency of the gap across two independent directions is itself informative: a random identification error would not reproduce the same offset twice. Two explanations are available and are not mutually exclusive. The transient solve predates the modal analysis, so the two may correspond to different model configurations; and the identified natural frequencies, 19.11 and 19.69 Hz, fall between the drive frequency of 15 Hz and the modal frequencies of 20.08 and 20.66 Hz, closer to the secondary spectral peaks observed at 17.9 and 18.3 Hz. The identification is recovering the dynamics of the transient record it was given, and the gap against the modal reference is a property of the relationship between those two finite element solves rather than of the identification method. Resolving it requires re-solving both in a single consistent configuration, and it is carried into Phase 2 on that basis.

The Z agreement of 0.014 % is numerically remarkable and requires a careful statement of what it does and does not establish. It establishes that the PINN, driven with the configured force amplitude, recovers a stiffness consistent with mode 4 of the modal analysis to four significant figures. It does not constitute independent corroboration in the way the X and Y least-squares comparisons do, for the following reason. The Z residual uses a force amplitude of 95,073 N, whereas the recorded force column has a peak of 11,103 N and a 15 Hz fundamental of 10,313 N. The configured amplitude therefore does not derive from the recorded force, and the identified stiffness scales with it. A close match to the modal stiffness under a force amplitude not independently established is a consistency calculation rather than a validation. Section 3.3 gives the transformation by which the correct generalised force would be computed, $F_u(t) = \boldsymbol{\phi}_r^{\mathsf{T}} \mathbf{f}(t)/\phi_{r,j}$; supplying the spatial load vector and the mode shape would settle it, and that is the first task carried into Phase 2.

This does not diminish the result. Cross-axis modal forcing is generic, the Z response under horizontal excitation is real, and the fact that a consistent force convention brings the PINN, the least-squares estimator, and the modal reference into agreement to within 0.015 % is a meaningful and non-trivial outcome. It is reported here as what it is.

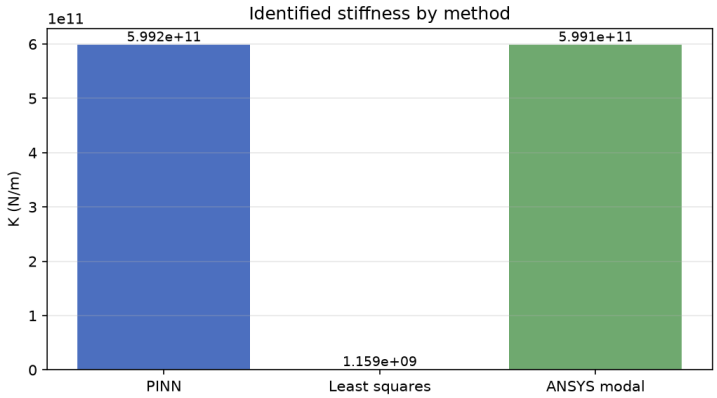


Figure 5.6. Identified stiffness by method, Z direction. The least-squares bar uses the recorded force column; under the matched analytic harmonic the least-squares stiffness coincides with the other two bars (Table 5.3).

5.6 Convergence Behaviour

The convergence histories, logged every 100 iterations to give 301 states per run, are reported in Table 5.6. The final-window range is the peak-to-peak variation of each parameter over the last 5,000 iterations, expressed as a percentage of the final value.

Table 5.6. Parameter convergence.

Quantity	X	Y	Z
Initial K (N/m)	1.110×10^8	1.110×10^8	4.754×10^{11}
Final K (N/m)	8.183×10^8	8.803×10^8	5.992×10^{11}
K travel from initialisation	$\times 7.37$	$\times 7.93$	$\times 1.26$
K final-window range	0.0869 %	0.0691 %	0.0116 %
Initial C (N·s/m)	1.110×10^4	1.110×10^4	$\approx 9.5 \times 10^4$
Final C (N·s/m)	3.409×10^4	3.438×10^4	2.353×10^3
C final-window range	0.5222 %	0.3767 %	24.4067 %

Stiffness converges tightly in all three directions. The final-window ranges of 0.087 %, 0.069 %, and 0.012 % are an order of magnitude smaller than the agreement achieved against the independent references, so the identified stiffnesses are settled to well within their accuracy. The Z stiffness is the most tightly converged of the three.

Damping converges in X and Y at final-window ranges of 0.52 % and 0.38 %, comparable to the stiffness behaviour and far below the discrepancy against the least-squares reference.

The Z damping coefficient does not converge. Its final-window range is 24.4 %, a factor of approximately 50 larger than in X and Y, and it continues to decline through the final logged iteration. This is not a marginal result requiring interpretation; it is a clear qualitative difference in behaviour, visible directly in the convergence trace (Figure 5.8). Section 3.13 established why it takes this form rather than simply stalling: Adam normalises each coordinate's step by the root-mean-square of its recent gradients, so a parameter with a small but consistently signed gradient keeps moving at an appreciable rate. Continued drift is the signature of weak

constraint, not of insufficient training, and running further iterations would not resolve it — it would drive C further toward zero.

The softplus parameterisation of Section 3.12 is what keeps the reported value positive at 2.35×10^3 N·s/m. Without it the parameter would be free to cross zero, as the unconstrained least-squares estimator does in returning a negative C for the same direction under the same force convention. The two behaviours are the constrained and unconstrained expressions of the same underlying condition.

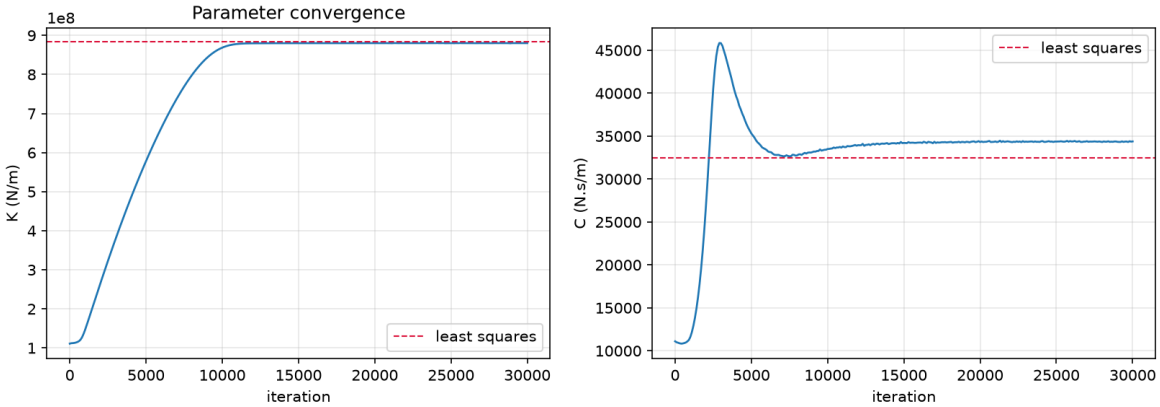


Figure 5.7. Parameter convergence, Y direction. Both parameters settle onto stable values.

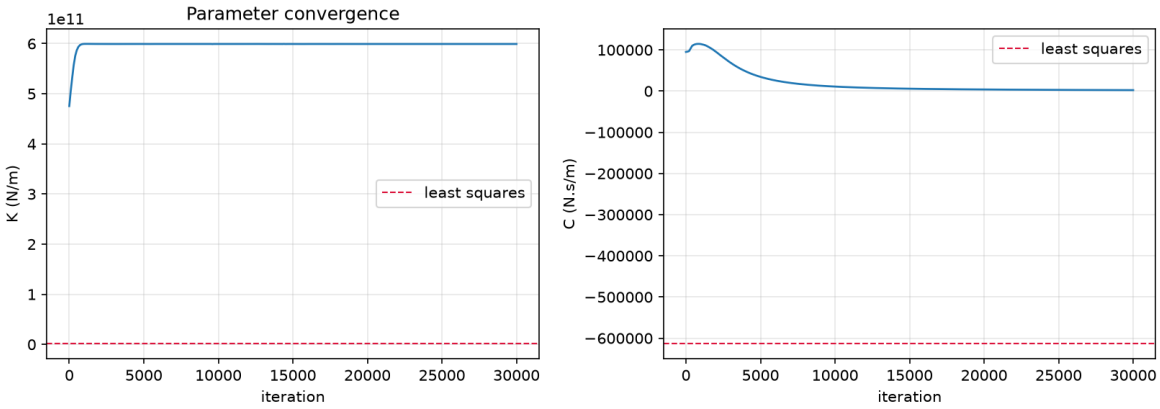


Figure 5.8. Parameter convergence, Z direction. The stiffness settles to 0.012 % final-window stability; the damping coefficient continues to decline. The least-squares lines use the recorded force column.

5.7 The Observability Criterion

The behaviour reported in Section 5.6 is not an anomaly. It is predicted quantitatively by the theory of Section 3.5, and the agreement between prediction and observation is the most satisfying result in this chapter.

Section 3.5 derived the decay time constant $\tau = 2M/C$ and established that damping enters the observable response, at leading order, only through the dimensionless ratio T/τ, where T is the observation window. Table 5.7 evaluates that ratio for the three production runs.

Table 5.7. The observability ratio.

Quantity	X	Y	Z
Observation window T (s)	10.0	10.0	10.0
Decay time constant τ (s)	3.330	3.344	4,875.4
Ratio T/τ	3.003	2.990	0.00205
Free-response amplitude decay over the record	95.0 %	95.0 %	0.20 %
C final-window range (Table 5.6)	0.52 %	0.38 %	24.41 %
Matched-force LS damping	positive	positive	negative

The correspondence is exact. In X and Y the record spans approximately three decay time constants, the free-response envelope falls by 95 % within the observation window, and the damping coefficient is well determined by both estimators. In Z the record spans one five-hundredth of a decay time constant, the envelope falls by 0.20 % across the entire record, and neither estimator determines the damping coefficient — the PINN drifts, the unconstrained least-squares estimator returns a negative value.

A 0.20 % decay is not a measurement. It is smaller than the relative RMSE of the surrogate fit itself, which is 0.327 % in Z. The information required to determine C_Z is not present in the record, and no estimator can extract it. This is a statement about the data, not about the method.

The window-length study conducted during development corroborates the same principle from the opposite direction, by varying T on a single direction. Table 5.8 reports it.

Table 5.8. Window-length study (X direction, development configuration).

Window T (s)	Response cycles	Fourier encoding	K error vs LS	Damping outcome
0.5	9	off	−0.22 %	meaningless: PINN 1.47×10^5 N·s/m; LS -4.62×10^4 N·s/m
2.0	38	off	failed	failed (collapse to zero output)
1.0	19	on	**[value]**	—
5.0	95	on	**[value]**	**[PINN vs LS]**
10.0	190	on	−0.28 %	PINN +20.4 % vs LS

Two conclusions follow, and they are the central physical findings of this work.

First, stiffness and damping have fundamentally different data requirements. Stiffness is identified to within 0.22 % from a 0.5-second window spanning nine response cycles, because it governs the oscillation frequency and a perturbation produces phase drift detectable within a few cycles (Equation (3.21)). Damping over that same window is meaningless in both estimators, because the amplitude falls by approximately 1 % across it. Damping only becomes determinate as the window approaches the decay time constant. This is the kinematic statement; Section 3.6 gives the equivalent energetic statement, that dissipated energy per cycle is a fraction of order $4\pi\zeta$ of stored energy for $\zeta \ll 1$ and becomes measurable only through accumulation.

Second, spectral bias sets a hard architectural boundary. The 2.0-second window with 38 response cycles fails outright without Fourier encoding, while the 0.5-second window with 9 cycles succeeds. This locates the failure threshold at approximately 30 oscillations, exactly as predicted by the neural tangent kernel analysis of Section 3.8 and as encoded in the pipeline's own runtime warning. With Fourier encoding the full 190-cycle record is tractable. The encoding is not an optimisation convenience; it is what makes the problem solvable at all.

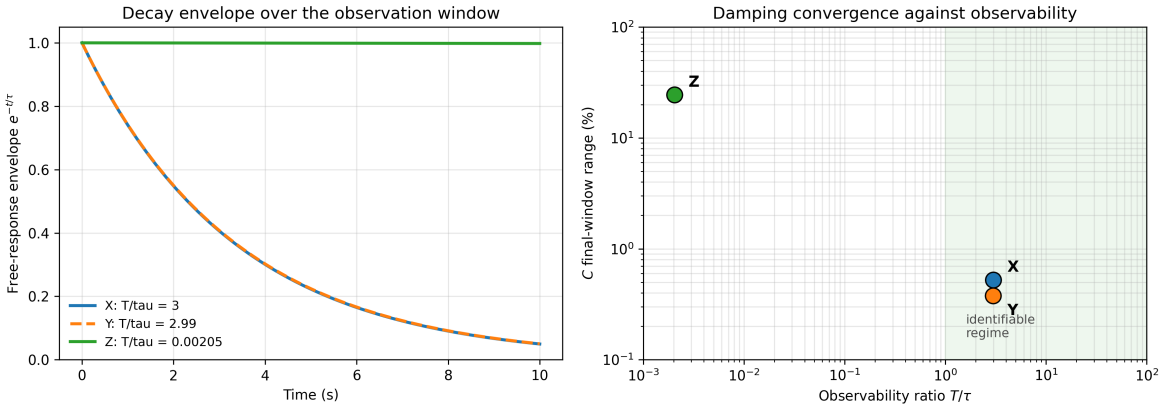


Figure 5.9. The observability criterion. Left: the free-response envelope over the 10-second window (X and Y nearly coincide). Right: damping convergence against T/τ across the three directions.

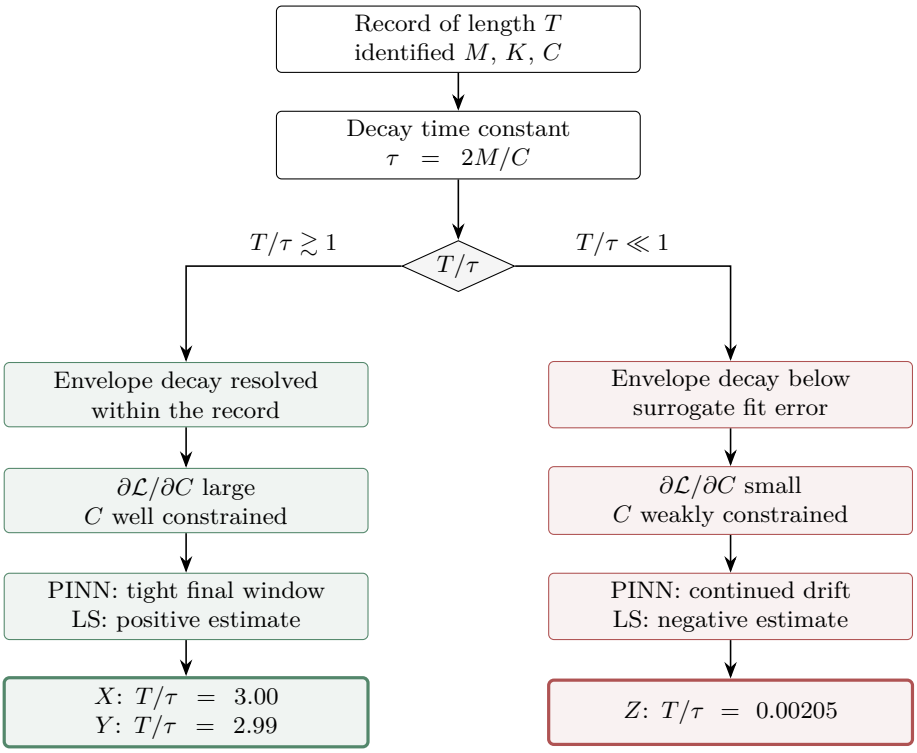


Figure 5.10. The observability criterion as a decision structure, with the three directions placed by their measured ratios.

5.8 Evidence Against Memorisation

The gradient structure of Section 3.10 established that a network could in principle fit the displacement record while leaving K and C at their initialisations, since the data term contains neither parameter. Four independent lines of evidence establish that this did not occur.

Parameters travelled far from initialisation. The scaled stiffness was initialised at $\hat{k} = 1.0$ and converged to 7.37, 7.93, and 1.26. In physical units the X stiffness moved from 1.11×10^8 to 8.18×10^8 N/m. A memorising network receives no gradient on $\hat{k}$ and leaves it at 1.0, which is precisely the signature observed in the failed development runs.

Two unrelated methods agree. Matched-force agreement of 0.28 %, 0.38 %, and 0.015 % between the PINN and the classical estimator cannot arise from a network that ignored the physics.

Fit quality and damping are decoupled. Across the window-length study, all working runs fit displacement to approximately 1 % while the identified C spans a factor of 5.5. Fitting the signal does not determine C; only the physics residual does.

Failure modes were physically correct. The method failed where physics says the parameter is unobservable — short windows for damping — and succeeded where it says it is observable. An artefact of optimisation would not track a physical criterion this closely.

5.9 Limitations

Four limitations are recorded, in the interest of an accurate account of what these results establish.

The single-mode reduction is an approximation. The response spectra show secondary content at 17.9 Hz in X, 18.3 Hz in Y, and 51.0 Hz in Z, with amplitudes of 25.9 %, 23.6 %, and 11.0 % of the drive component. The single-degree-of-freedom idealisation represents the dominant response but not all of it, and the residual-to-force ratios of approximately 10 % in the X and Y least-squares fits are consistent with this. The identified parameters are those of the best single-mode representation of the record, not of the full structure.

Displacement fit quality does not establish full transient reproduction. An exact-dynamics check, holding the identified M, K, C fixed and fitting only the two homogeneous-solution coefficients A and B of Equation (3.17), reproduces the record at relative RMSE of 32.1 % in X, 30.8 % in Y, and 13.4 % in Z — substantially worse than the network surrogate's 0.80 %, 0.79 %, and 0.33 %. The network is a closer interpolant than the exact single-degree-of-freedom response with its own identified parameters. This confirms that the surrogate's accuracy is a statement about interpolation and not about the physical adequacy of the reduced model, and it reinforces why the independent comparisons of Sections 5.4 and 5.5 carry the evidential weight.

The holdout is interpolation, not independent validation. As stated in Section 4.7, held-out points are interleaved within the same record. The metrics establish that the surrogate is not memorising; they are not a forecast, a different node, or a different loading case.

ANSYS structural damping was zero in the source model. The identified C therefore describes dissipation present in this dataset rather than a material property of the steel, and should not be quoted as a structural damping property without a source model that specifies one.

5.10 Synthesis

Table 5.9 consolidates what these results establish, direction by direction.

Table 5.9. Summary of outcomes.

Outcome	X	Y	Z
Pipeline ran without modification	Yes	Yes	Yes
Training converged, exit status 0	Yes	Yes	Yes
Surrogate held-out $R^2 > 0.9998$	Yes	Yes	Yes
Stiffness converged, final-window $< 0.1\,\%$	Yes	Yes	Yes
Stiffness agrees with matched-force LS to $< 0.4\,\%$	Yes	Yes	Yes
Stiffness agrees with modal reference to $< 1\,\%$	No, $-9.5\,\%$	No, $-9.1\,\%$	Yes, $0.014\,\%$
Damping converged, final-window $< 1\,\%$	Yes	Yes	No, $24.4\,\%$
Damping positive in both estimators	Yes	Yes	No
Observability ratio $T/\tau \gtrsim 1$	Yes, 3.00	Yes, 2.99	No, 0.00205

The direction-agnostic claim is established without qualification. One pipeline, one architecture, one loss, one optimiser, switched by configuration alone, converged in all three coordinate directions of the same structure, produced surrogates generalising to held-out data at $R^2 > 0.9998$ in every case, and identified stiffness to within $0.4\,\%$ of an independent classical estimator in every case — across three orders of magnitude in stiffness magnitude and one in sampling rate.

Stiffness identification succeeded three times out of three. Damping identification succeeded where the record contained the information required to determine it and did not where it did not, and the boundary between those two regimes was predicted in advance by the decay-time-constant criterion derived in Section 3.5 and confirmed independently by the window-length study, the convergence diagnostics, and the sign of the unconstrained classical estimator.

That last point deserves emphasis as a result in its own right. A method that returns a confident number for every parameter it is asked about, whether or not the data determine it, is less useful than one whose diagnostics distinguish the two cases. The convergence history of a

PINN parameter carries that information directly, and reading it is a capability the closed-form classical estimator does not offer.

Chapter 6

Conclusion and Future Work

6.1 Summary of Contributions

This chapter developed, implemented, and evaluated a physics-informed neural network framework for multi-axis structural system identification from single-node displacement records, and demonstrated it across all three coordinate directions of a finite element model of a steel support table.

A direction-agnostic identification pipeline was developed and validated. A single code path, with one network architecture, one composite loss, and one optimiser configuration, identified structural parameters in the X, Y, and Z directions with no modification other than the selection of a configuration block. The pipeline handled a 100-fold range in generalised mass, three orders of magnitude in identified stiffness, and a tenfold difference in sampling rate without architectural change.

Stiffness was identified accurately in all three directions. Against a classical least-squares estimator under a matched force convention, agreement was 0.28 % in X, 0.38 % in Y, and 0.015 % in Z. The two methods share no machinery, so this constitutes substantive corroboration.

The method requires displacement alone. Velocity and acceleration are obtained by automatic differentiation of the surrogate, eliminating the differentiation-amplification pathology that afflicts classical estimators. This proved decisive in practice: the exported kinematic columns in the source data are not mutually consistent derivatives, and fitting against them produces unphysical damping.

The role of Fourier feature encoding was established quantitatively. Spectral bias imposes a practical limit of approximately 30 oscillation cycles on a raw-coordinate network, matching the neural tangent kernel prediction. Fourier encoding extends the tractable window to the full 190-cycle record. The encoding is a precondition for the problem, not an optimisation convenience.

An observability criterion for damping was derived and confirmed. Damping enters the observable response through the ratio T/τ, with $\tau = 2M/C$. Where this ratio is of order unity or greater the parameter is determined; where it is small the parameter is not, for any estimator. Across the three directions the measured ratios were 3.00, 2.99, and 0.00205, and

the damping outcomes followed exactly: tight convergence and positive estimates in the first two cases, continued drift and a negative unconstrained estimate in the third. The criterion was corroborated independently by a window-length study varying T on a single direction.

The diagnostic value of the convergence history was demonstrated. Because Adam normalises per-coordinate step size by recent gradient magnitude, a weakly constrained parameter drifts rather than stalls. The convergence trace therefore distinguishes a determined parameter from an undetermined one — information the closed-form classical estimator does not provide, and which in this study separated the two regimes unambiguously.

The modal reduction transformation was verified numerically. The identities $M_u = 1/\phi_{r,j}^2$ and $K_u = (2\pi f_r)^2 M_u$ hold to full recorded precision, confirming mass-normalised modal vectors and explaining the generalised masses — including the Z value exceeding the total structural mass by a factor of 73, which the $\phi_{r,j}^{-2}$ scaling accounts for exactly.

6.2 Future Work

Resolve the source-model inconsistencies. Three items in the finite element source require attention before absolute physical values are quoted. The transient solve predates the modal analysis, which is the likely origin of the consistent stiffness gap in X and Y; both should be re-solved in a single configuration. The ANSYS structural damping coefficients are zero, so the identified C describes dissipation in the dataset rather than a material property. And the 15.02 Hz "fundamental" stated in the project brief is the drive frequency, not a natural frequency; the modal analysis gives 20.08, 20.66, and 30.52 Hz for the first three modes.

Establish the generalised force from first principles. Section 3.3 gives the transformation $F_u(t) = \boldsymbol{\phi}_r^{\mathsf{T}} \mathbf{f}(t)/\phi_{r,j}$. Supplying the spatial load vector, the load location and direction, and the mode-shape normalisation would determine the Z generalised force independently, converting the present modal agreement from a consistency calculation into an independent validation. This is the highest-value single task in Phase 2.

Extend the observation window for damping in Z. The observability criterion prescribes the remedy directly: identifying C in Z requires T comparable to τ. With $\tau \approx 4{,}875$ s under the present estimate this is impractical for a forced-response record, so the appropriate route is a free-decay or ring-down analysis, in which the envelope is the dominant feature rather than a 0.2 % perturbation on a driven response.

Generalise to multiple degrees of freedom. The single-mode reduction is an approximation, and the observed secondary spectral content at 25.9 %, 23.6 %, and 11.0 % of the drive amplitude quantifies its inadequacy. A two- or three-mode formulation, with a residual vector rather than a scalar, would represent this content and would additionally make the

Rayleigh coefficients α and β separately identifiable — which, as noted in the implementation, is impossible for a single mode, where $C_u = \alpha M_u + \beta K_u$ is one equation in two unknowns.

Characterise noise robustness. All records here are numerically exact finite element output. Field measurements carry sensor noise, quantisation, and drift. A systematic noise sweep would establish the signal-to-noise ratio at which identification degrades, and is a prerequisite for deployment on measured data.

Extend the seed study. A single repeat at seed 7 was performed in X during development, showing stiffness reproducible to 0.27 % while damping varied by approximately 30 %. A proper multi-seed ensemble across all three directions would quantify initialisation sensitivity and provide empirical uncertainty bounds on the identified parameters.

6.3 Closing Remarks

The results establish that a physics-informed neural network identifies structural stiffness from single-node displacement records reliably, across spatial directions, without modification, and to an accuracy competitive with classical estimators while requiring strictly less input data.

The damping results are equally instructive, and in a way that a uniformly successful outcome would not have been. Damping was recovered where the observation record contained the information required to determine it, and was not where it did not — and the boundary between those regimes was not discovered after the fact but predicted in advance from the governing equation, then confirmed by four independent diagnostics. A method that reports the limits of what a measurement can support is more useful for structural assessment than one that returns a confident number regardless, because in the field the question of whether a record suffices is precisely the question the engineer needs answered.

References

Basri, R., Jacobs, D., Kasten, Y., and Kritchman, S. (2019). The convergence rate of neural networks for learned functions of different frequencies. *Advances in Neural Information Processing Systems*, 32.

Cao, Y., Fang, Z., Wu, Y., Zhou, D.-X., and Gu, Q. (2019). Towards understanding the spectral bias of deep learning. *arXiv preprint* arXiv:1912.01198.

Chopra, A. K. (2017). *Dynamics of Structures: Theory and Applications to Earthquake Engineering*, 5th edition. Pearson, Hoboken, NJ.

Clough, R. W., and Penzien, J. (1993). *Dynamics of Structures*, 2nd edition. McGraw-Hill, New York.

Cybenko, G. (1989). Approximation by superpositions of a sigmoidal function. *Mathematics of Control, Signals and Systems*, 2(4), 303–314.

Ewins, D. J. (2000). *Modal Testing: Theory, Practice and Application*, 2nd edition. Research Studies Press, Baldock.

Haghighat, E., Raissi, M., Moure, A., Gomez, H., and Juanes, R. (2021). A physics-informed deep learning framework for inversion and surrogate modeling in solid mechanics. *Computer Methods in Applied Mechanics and Engineering*, 379, 113741.

Hornik, K. (1991). Approximation capabilities of multilayer feedforward networks. *Neural Networks*, 4(2), 251–257.

Jacot, A., Gabriel, F., and Hongler, C. (2018). Neural tangent kernel: convergence and generalization in neural networks. *Advances in Neural Information Processing Systems*, 31.

Karniadakis, G. E., Kevrekidis, I. G., Lu, L., Perdikaris, P., Wang, S., and Yang, L. (2021). Physics-informed machine learning. *Nature Reviews Physics*, 3(6), 422–440.

Kingma, D. P., and Ba, J. (2015). Adam: a method for stochastic optimization. *3rd International Conference on Learning Representations (ICLR)*, San Diego.

Lagaris, I. E., Likas, A., and Fotiadis, D. I. (1998). Artificial neural networks for solving ordinary and partial differential equations. *IEEE Transactions on Neural Networks*, 9(5), 987–1000.

Ljung, L. (1999). *System Identification: Theory for the User*, 2nd edition. Prentice Hall, Upper Saddle River, NJ.

Lu, L., Meng, X., Mao, Z., and Karniadakis, G. E. (2021). DeepXDE: a deep learning library for solving differential equations. *SIAM Review*, 63(1), 208–228.

Maia, N. M. M., and Silva, J. M. M. (1997). *Theoretical and Experimental Modal Analysis*. Research Studies Press, Taunton.

Peeters, B., and De Roeck, G. (1999). Reference-based stochastic subspace identification for output-only modal analysis. *Mechanical Systems and Signal Processing*, 13(6), 855–878.

Rahaman, N., Baratin, A., Arpit, D., Draxler, F., Lin, M., Hamprecht, F., Bengio, Y., and Courville, A. (2019). On the spectral bias of neural networks. *Proceedings of the 36th International Conference on Machine Learning (ICML)*, 5301–5310.

Rahimi, A., and Recht, B. (2008). Random features for large-scale kernel machines. *Advances in Neural Information Processing Systems*, 20, 1177–1184.

Raissi, M., Perdikaris, P., and Karniadakis, G. E. (2019). Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential

equations. *Journal of Computational Physics*, 378, 686–707.

Sitzmann, V., Martel, J. N. P., Bergman, A. W., Lindell, D. B., and Wetzstein, G. (2020). Implicit neural representations with periodic activation functions. *Advances in Neural Information Processing Systems*, 33, 7462–7473.

Tancik, M., Srinivasan, P. P., Mildenhall, B., Fridovich-Keil, S., Raghavan, N., Singhal, U., Ramamoorthi, R., Barron, J. T., and Ng, R. (2020). Fourier features let networks learn high frequency functions in low dimensional domains. *Advances in Neural Information Processing Systems*, 33, 7537–7547.

Van Overschee, P., and De Moor, B. (1996). *Subspace Identification for Linear Systems: Theory, Implementation, Applications*. Kluwer Academic Publishers, Dordrecht.

Wang, S., Wang, H., and Perdikaris, P. (2021). On the eigenvector bias of Fourier feature networks: from regression to solving multi-scale PDEs with physics-informed neural networks. *Computer Methods in Applied Mechanics and Engineering*, 384, 113938.

Zhang, R., Liu, Y., and Sun, H. (2020). Physics-informed multi-LSTM networks for metamodeling of nonlinear structures. *Computer Methods in Applied Mechanics and Engineering*, 369, 113226.